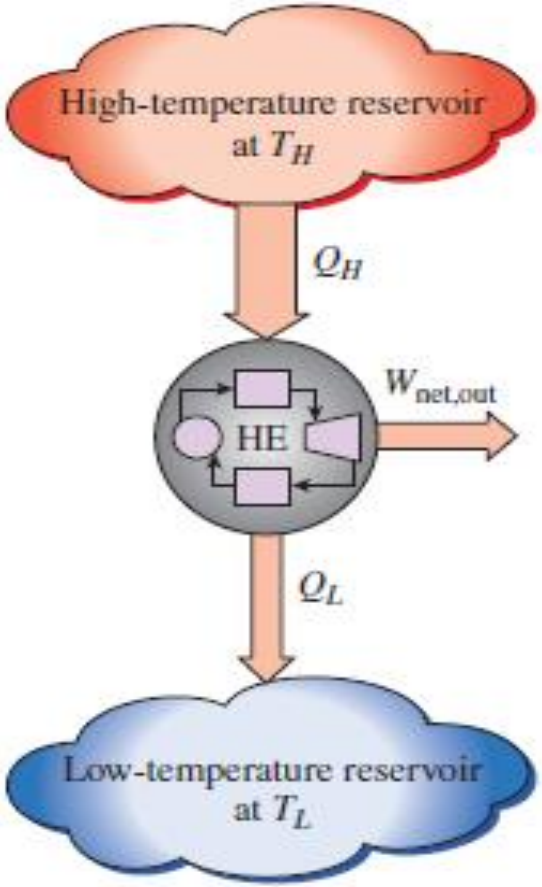
ENGINEERING THERMODYNAMICS (MME 365)

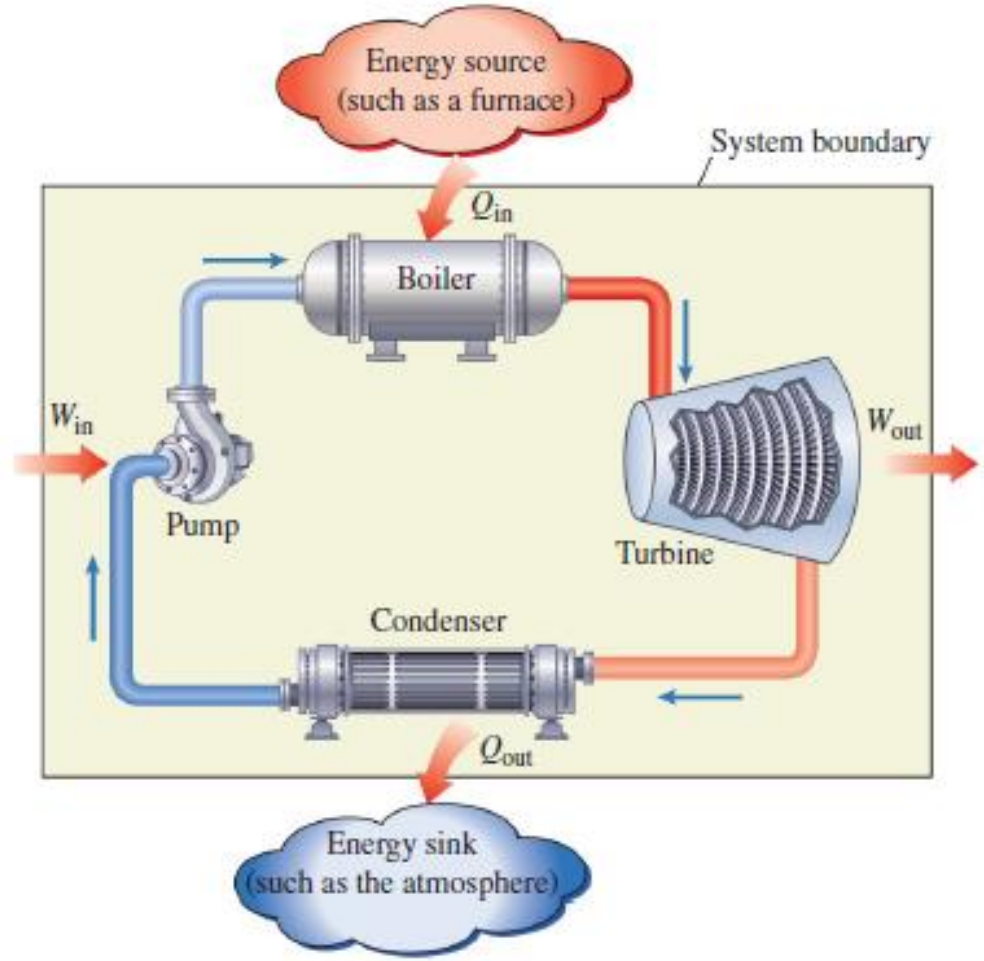
VAPOUR POWER CYCLES

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Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Vapour Power Cycles

Heat Engine Cycle



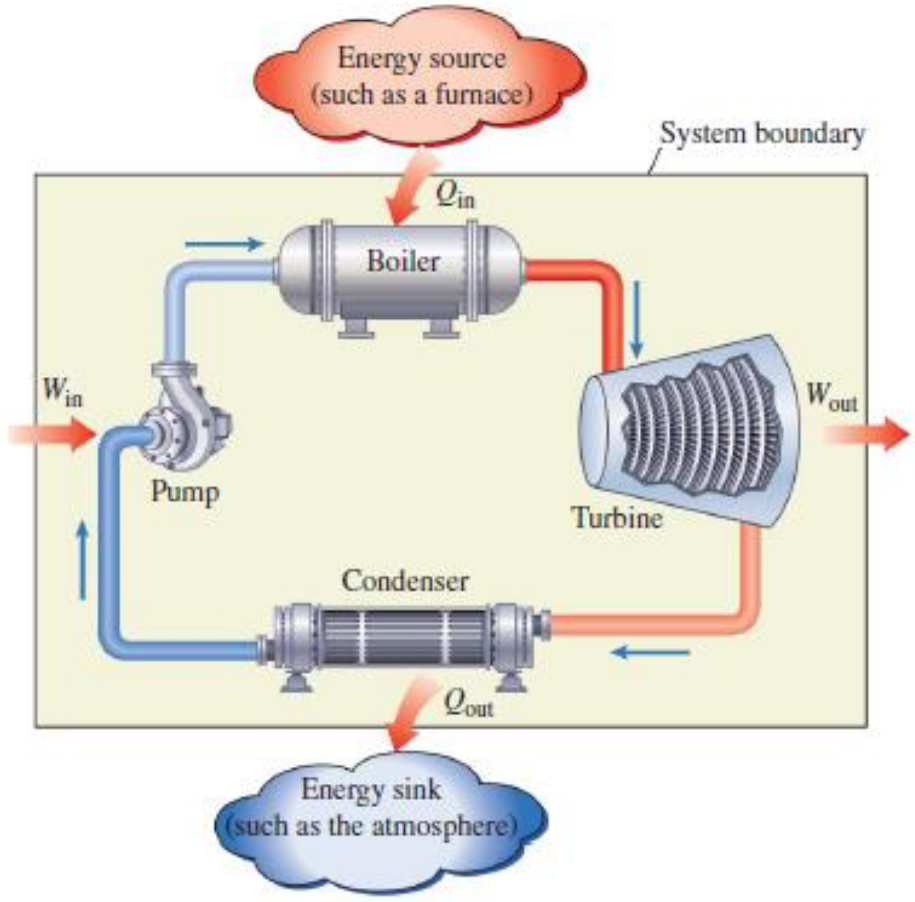


Steam power plant

Vapour Power Cycles

Characteristics of a vapour power cycle

- The working fluid is a condensable vapour, which can be in the liquid phase during part of the cycle.
- The cycle consists of steady flow processes with each process carried out in a separate component specially designed for the purpose.
- Each component constitutes an open system and all the components are connected in series so that each fluid element passes through thermodynamic states.
- In each component, matter flows into and out of region of space. All the processes are considered to be flow processes.



Steam power plant

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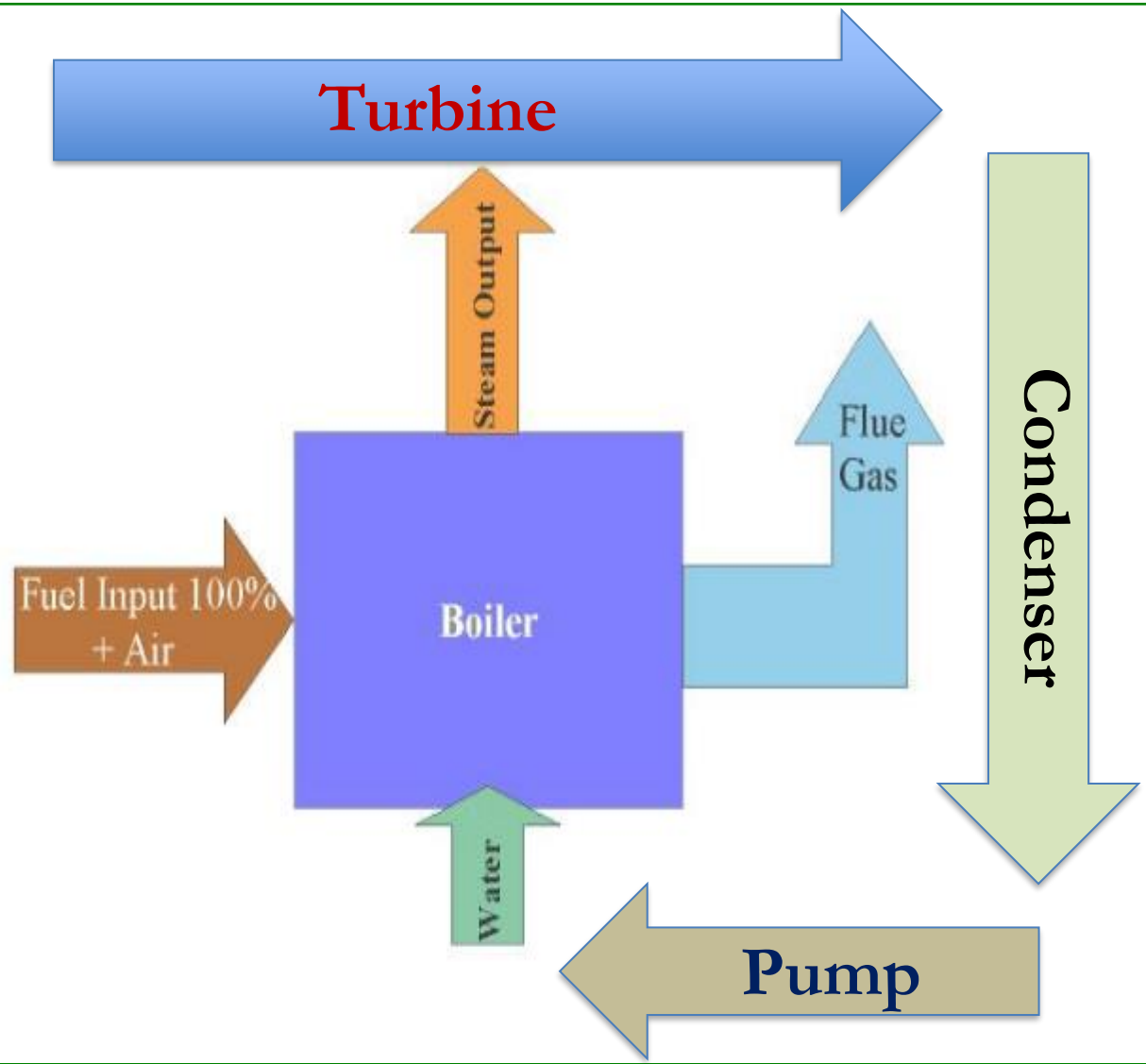
Criteria for Comparing Power Cycles

- The choice of a power plant is determined largely by:
- **Capital cost**: This is determined by size and complexity of the plant.
  - ✓ This is determined by the specific steam consumption, SSC. It is the mass flow of steam required per unit net power output.

$$\text{SSC} = {}^{3600}/_{W_{net}} \left({}^{kg}/_{kWh}\right)$$

Where W_{net} is in kJ/kg.

- **Operating cost:** This is determined by the overall efficiency of the plant and work ratio.
  - ✓ The work ratio is the ratio of net work output to the turbine work output. $r_w = {}^{W_{net}}/_{W_{turbine}}$
  - ✓ The overall efficiency is the ratio of net work output of the cycle to the fuel energy input to the boiler.

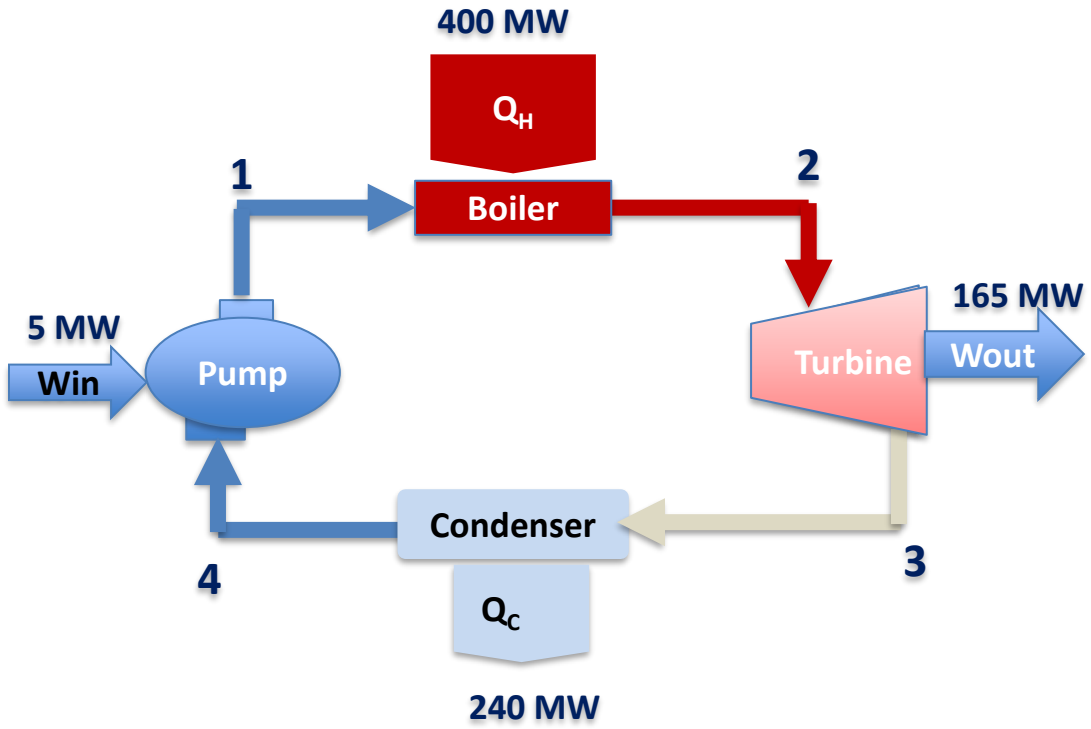


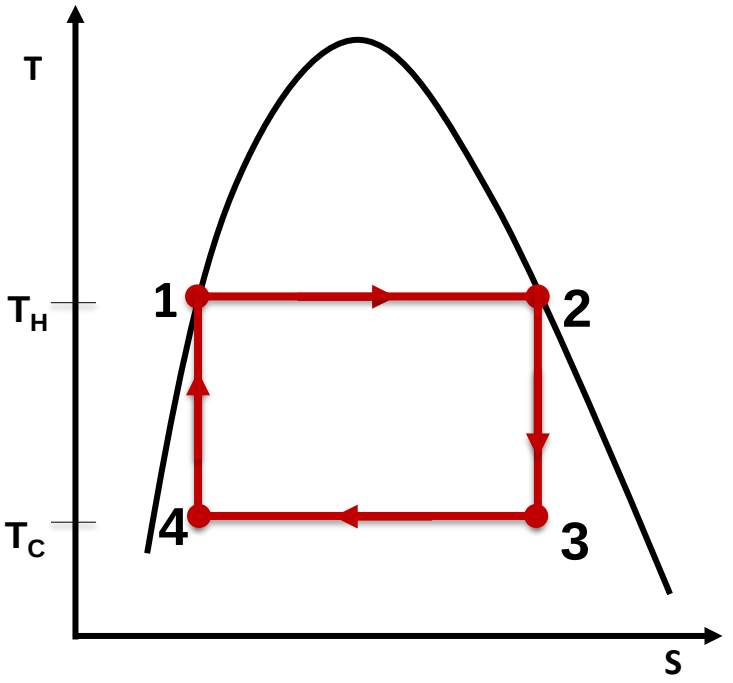
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Carnot Vapour Power Cycle

The Carnot Cycle consists of four (4) processes

- Two isothermal (constant temperature) heat transfer processes
- Two reversible adiabatic (constant entropy) work transfer processes





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Carnot Vapour Power Cycle

- **Each component is treated as an open system.**
- The steady flow energy equation for an open system from an initial state, i to a final state f, is given by

$$Q_{if} - W_{if} = m\left[(h_f - h_i) + \frac{C_f^2 - C_i^2}{2} + g(Z_f - Z_i)\right]$$

- **Applying the SFEE to the boiler,**

$$\boldsymbol{Q_{12} = m(h_2 - h_1) = mT_H(s_2 - s_1)}$$

- **Applying the SFEE to the turbine,**

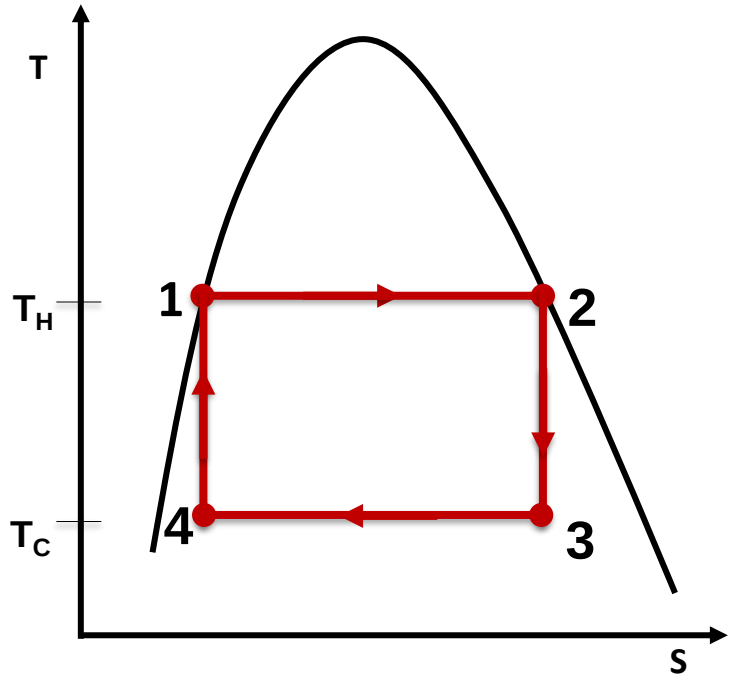
$$\boldsymbol{-W_{23} = m(h_3 - h_2)}$$

- **Applying the SFEE to the condenser,**

$$\boldsymbol{Q_{34} = m(h_4 - h_3) = mT_c(s_4 - s_3)}$$

- **Applying the SFEE to the pump,**

$$\boldsymbol{-W_{41} = m(h_1 - h_4) = mv_4(P_1 - P_4)}$$



$$\eta_{Carnot} = 1 - \frac{T_C}{T_H}$$

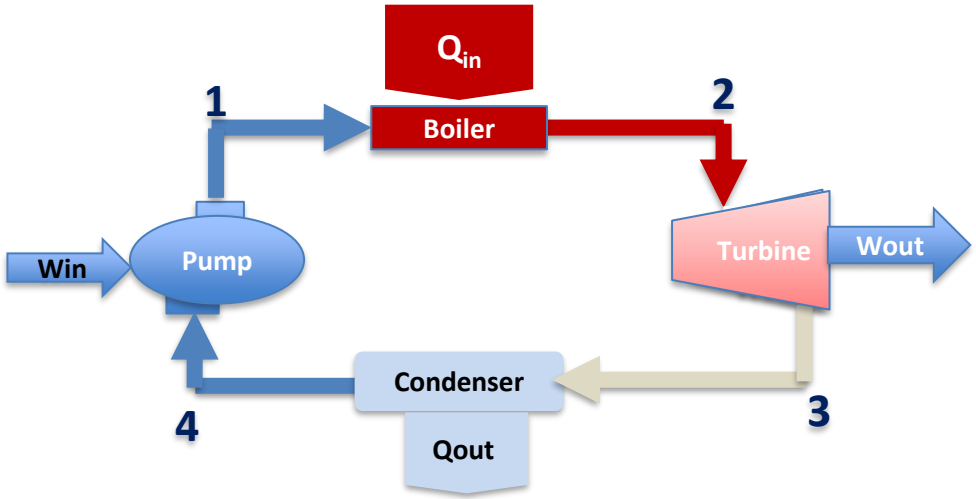
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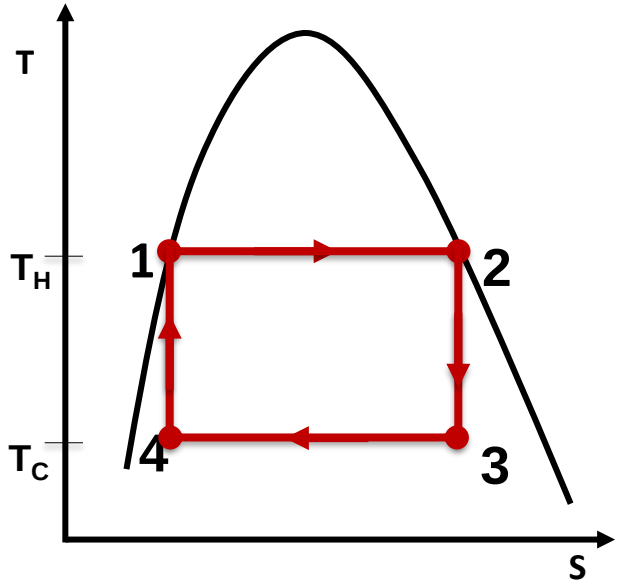
Carnot Power Cycle Problem

Problem

A steady-flow Carnot cycle uses water as the working fluid. Water changes from saturated liquid to saturated vapor as heat is transferred to it from a source at 250 °C. Heat rejection takes place at a pressure of 20 kPa. Show the cycle on a T-s diagram relative to the saturation lines, and determine (a) the thermal efficiency, (b) the amount of heat rejected, and (c) the net work output.

Suggested Answers: 36.3%; 1715.3 kJ/kg; 623 kJ/kg

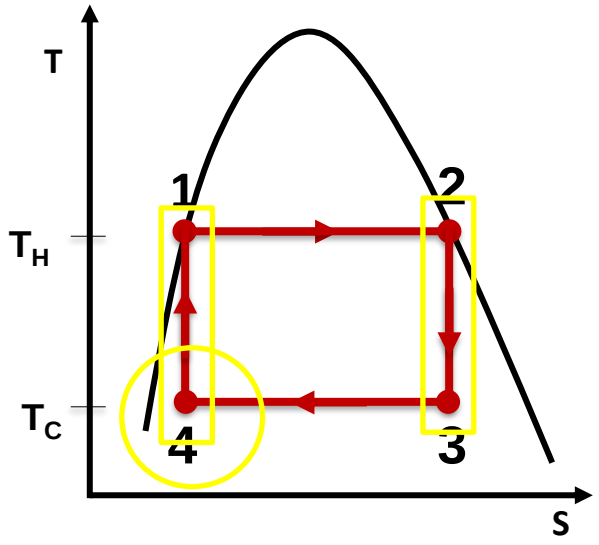




Limitations of the Carnot Cycle

Reasons why the Carnot cycle is not a practical cycle

- The turbine has to handle vapour with low quality: risk of erosion and wear in turbine blades.
- It is impractical to design a pump that handles two phase fluid.
- It is difficult to control the condensation process that precisely has to end up with the desired quality at point 4.

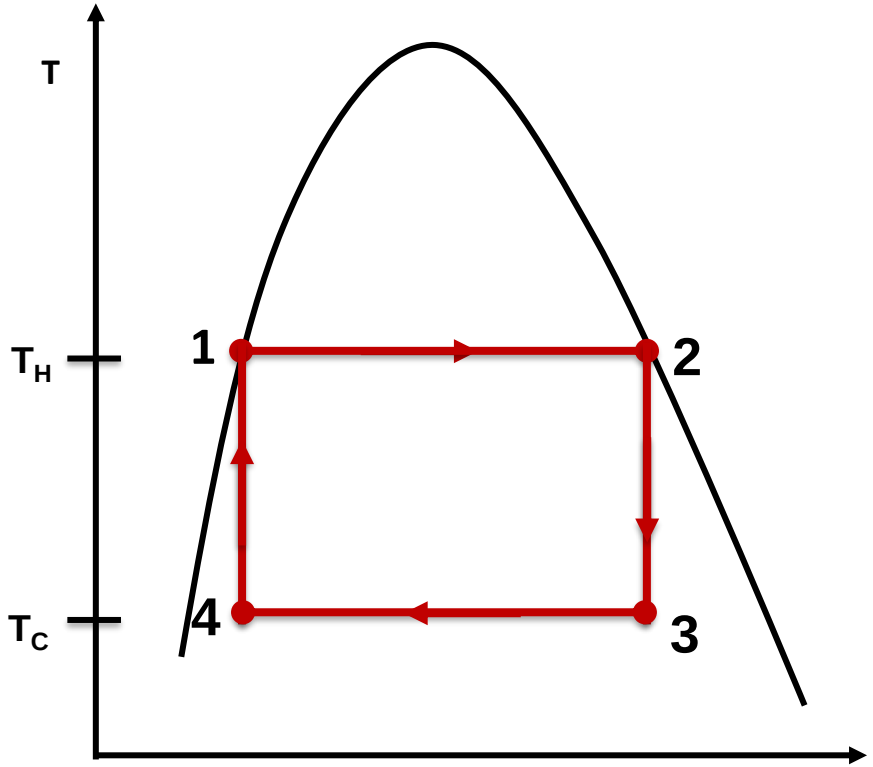


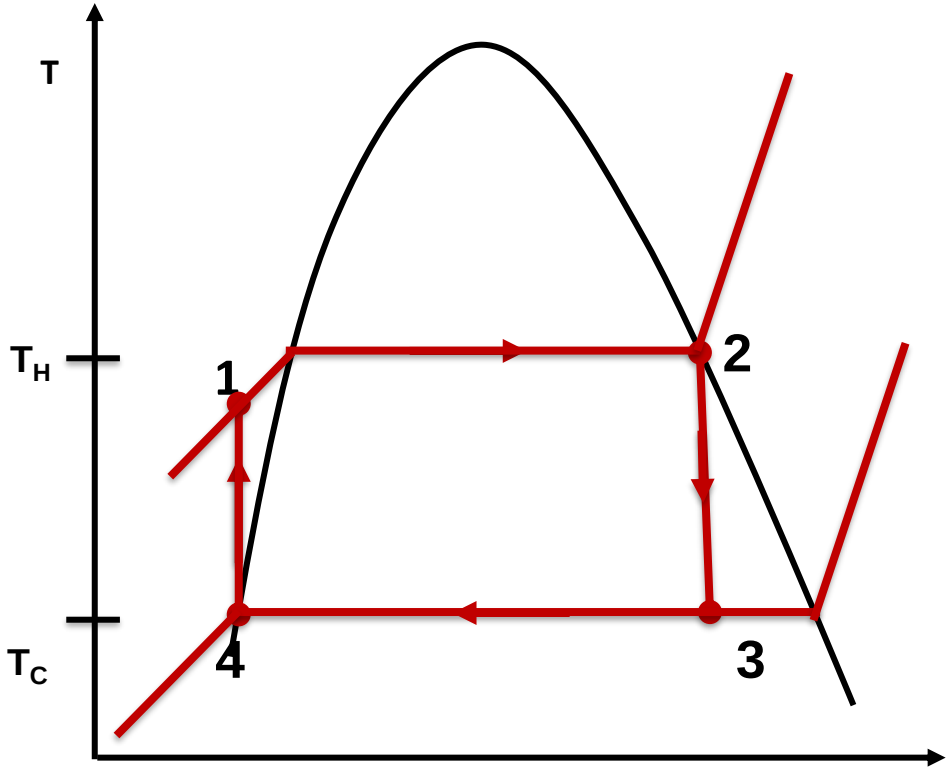
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Carnot And Ideal Rankine Cycles

Carnot cycle :
- Two isothermal processes
- Two isentropic processes

Ideal Simple Rankine cycle :
- Two isobaric processes
- Two isentropic processes





Ideal Rankine Vapour Power Cycle

- **Each component is treated as an open system.**
- The steady flow energy equation for an open system from an initial state, i to a final state f, is given by

$$Q_{if} - W_{if} = m\left[(h_f - h_i) + \frac{C_f^2 - C_i^2}{2} + g(Z_f - Z_i)\right]$$

- **Applying the SFEE to the boiler,**

$$Q_{12} = m(h_2 - h_1)$$

- **Applying the SFEE to the turbine,**

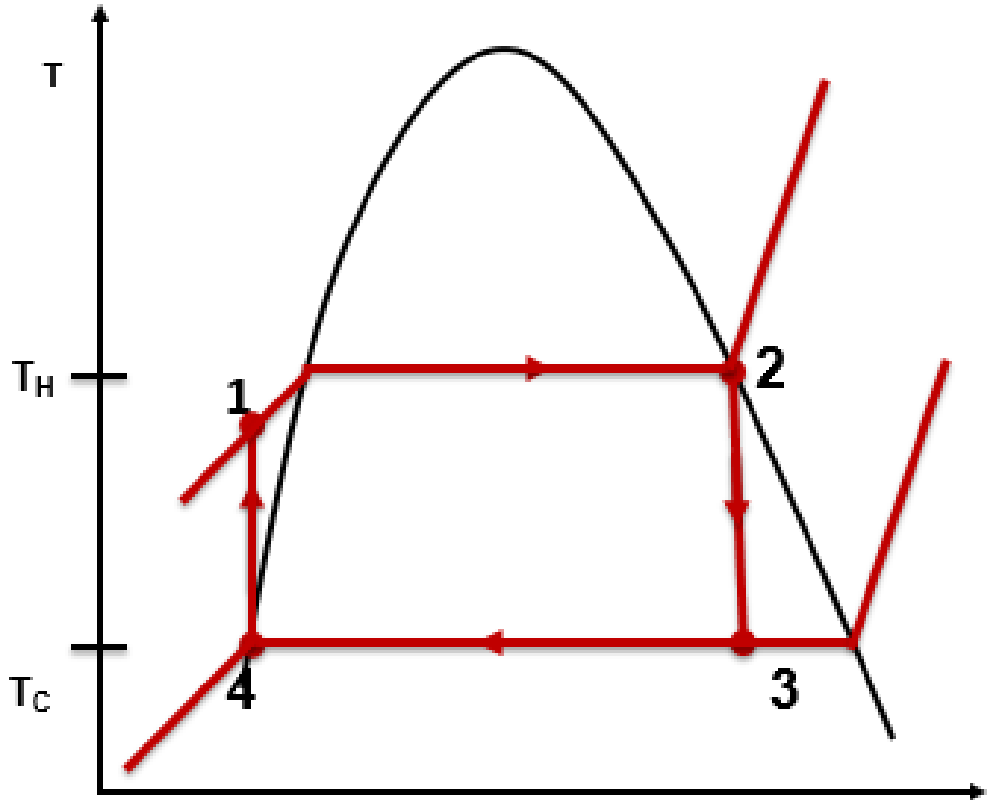
$$-W_{23} = m(h_3 - h_2)$$

- **Applying the SFEE to the condenser,**

$$Q_{34} = m(h_4 - h_3)$$

- **Applying the SFEE to the pump,**

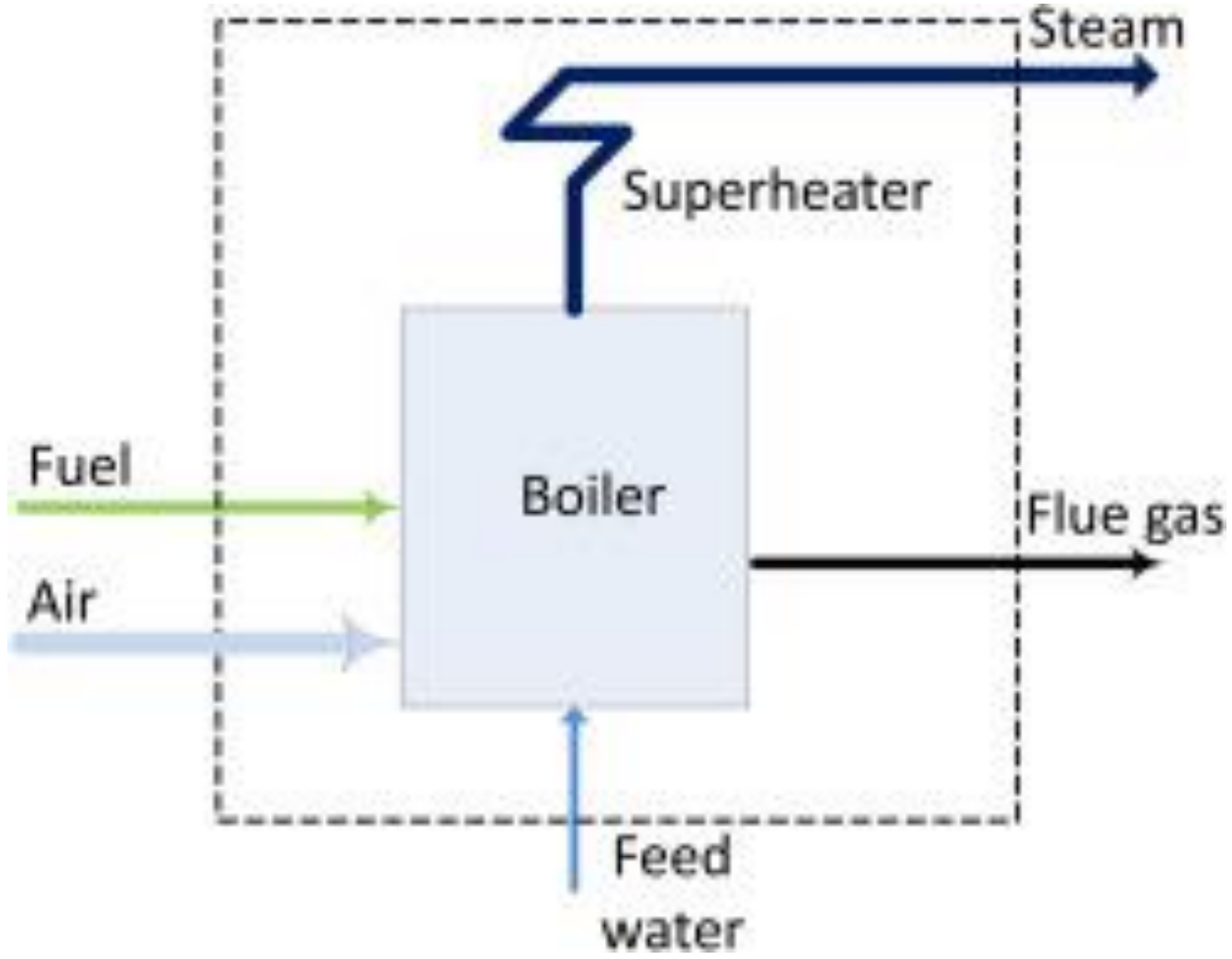
$$-W_{41} = m(h_1 - h_4) = mv_4(P_1 - P_4)$$



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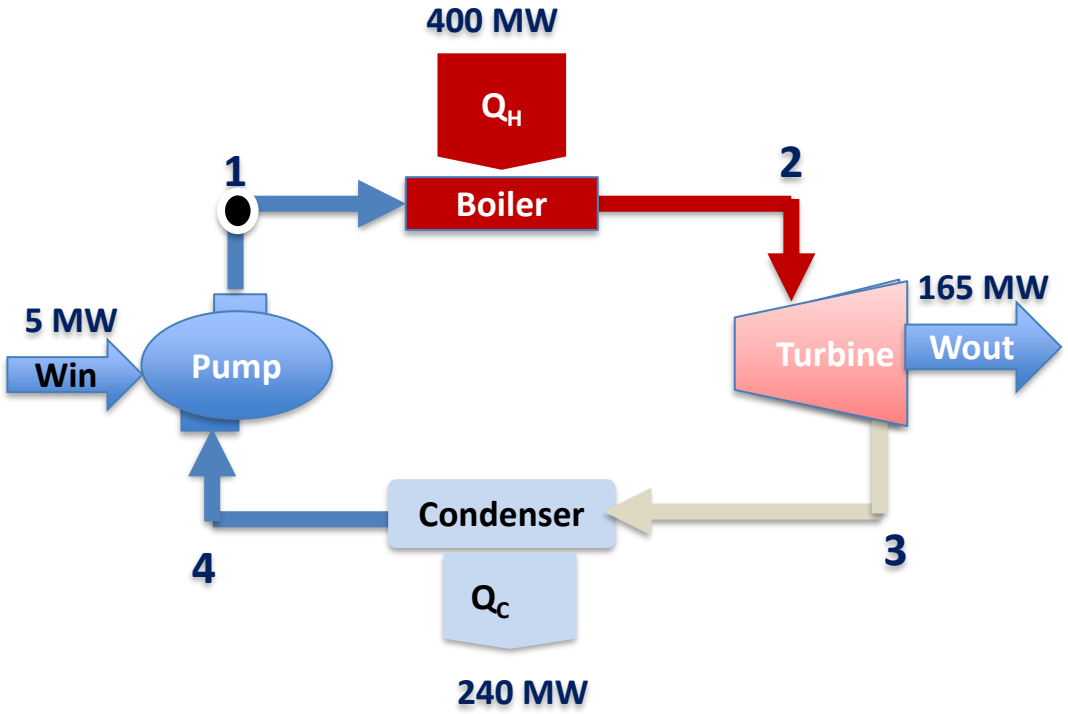
Improvements In The Ideal Rankine Cycle

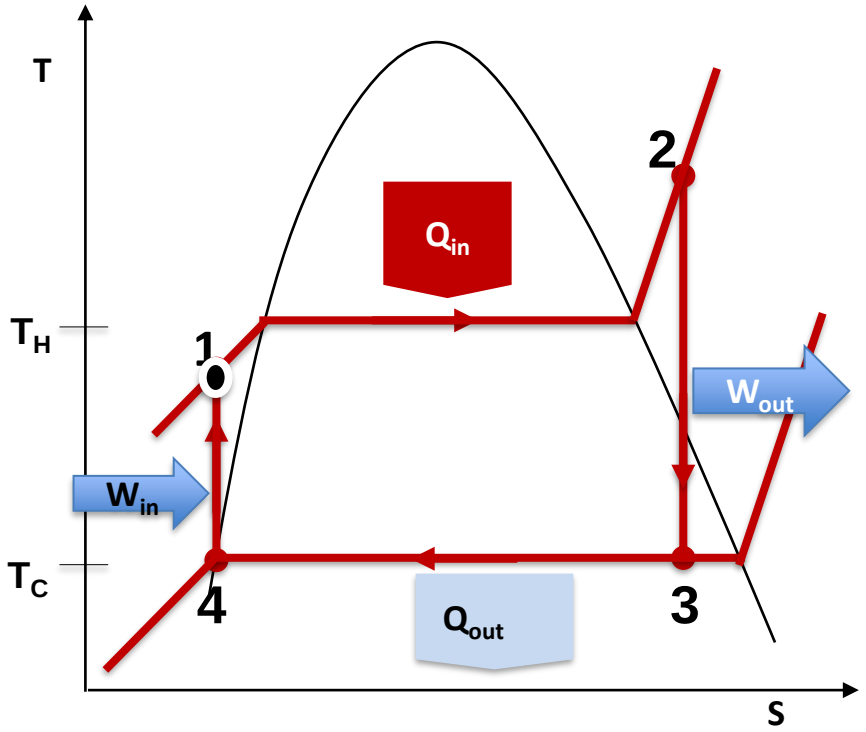
- In general, the efficiency can be improved by increasing the complexity of the plant.
- A compromise between low operating and capital costs is therefore required.
- One way to improve the efficiency of the Rankine cycle is by superheating the saturated steam before it enters the turbine.
- This can be done by installing a superheater in the steam flow line at exit from the boiler.



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Improvements In The Ideal Rankine Cycle





$$W_{net,out} = W_{out} - W_{in}$$

$$Q_{net} = Q_H - Q_C$$

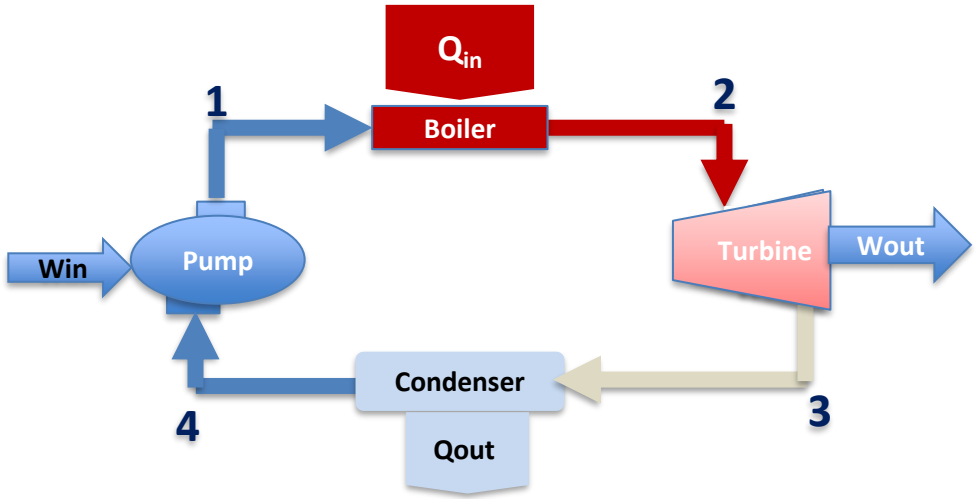
$$Q_{net} = W_{net}$$

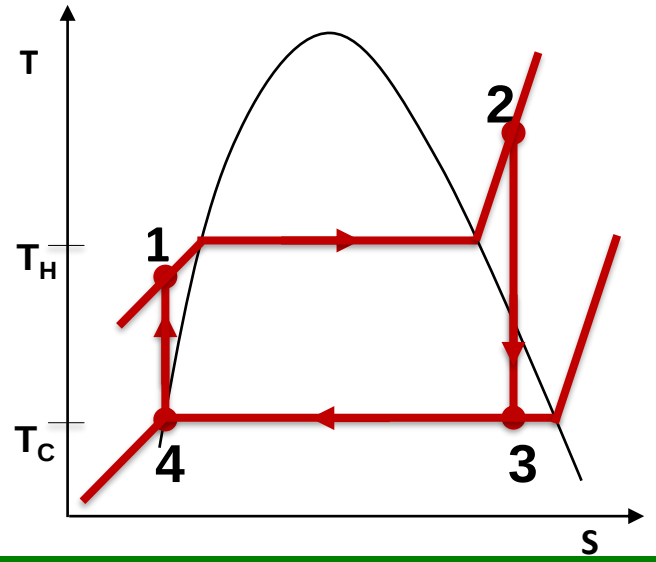
Ideal Rankine Cycle with Superheat Problem

Problem 1

Consider a 210-MW steam power plant that operates on a simple ideal Rankine cycle. Superheated steam enters the turbine at 10 MPa and 500 °C and is cooled in the condenser at a pressure of 10 kPa. Show the cycle on a T-s diagram with respect to saturation lines, and determine (a) the quality of the steam at the turbine exit, (b) the thermal efficiency of the cycle, and (c) the mass flow rate of the steam.

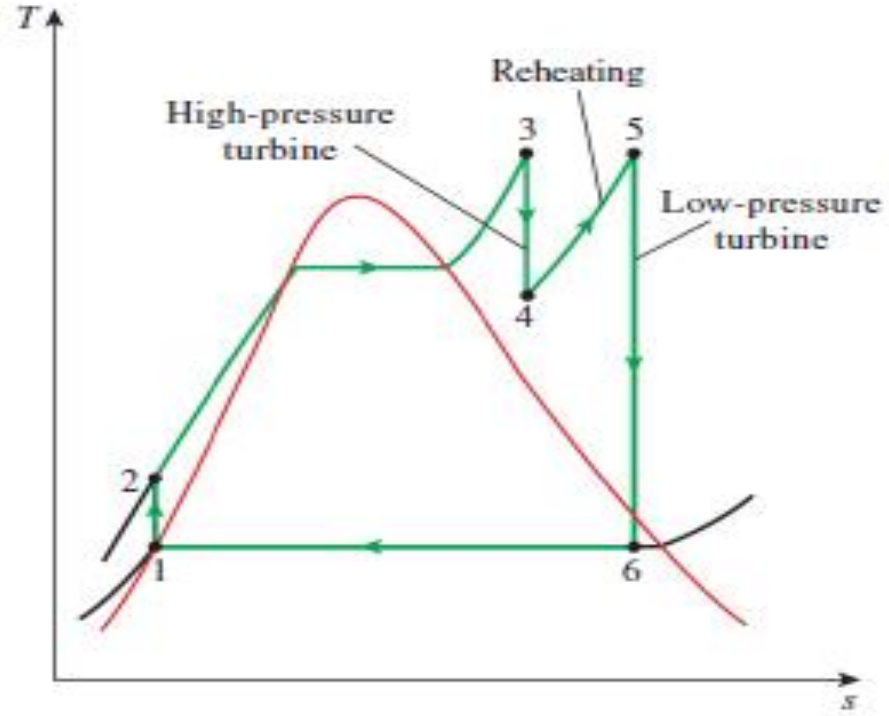
Suggested Answers: (a) 0.793, (b) 40.2 percent, (c) 165 kg/s

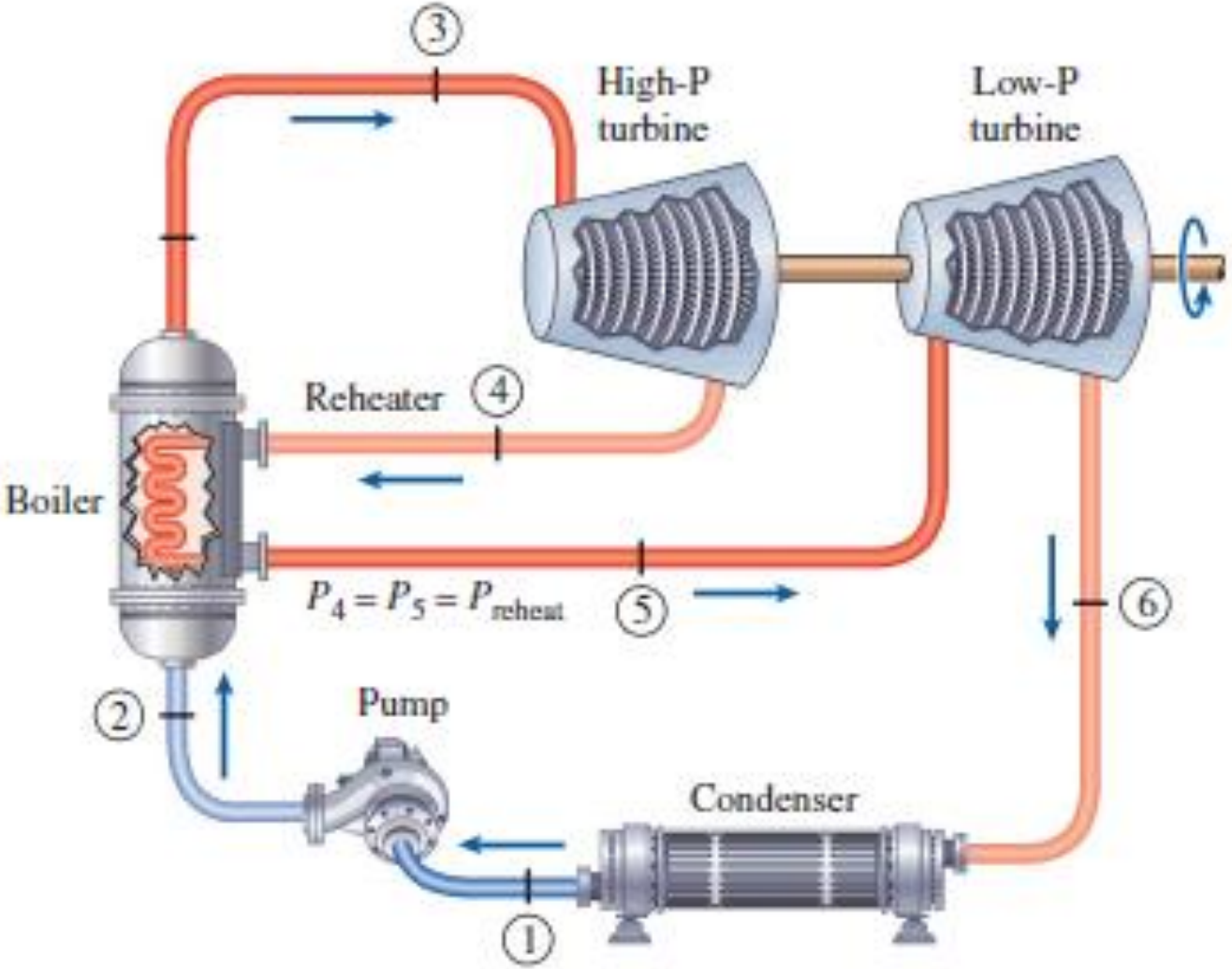




Improvements In The Ideal Rankine Cycle

- Modify the simple ideal Rankine cycle with a reheat process.
- **Sole purpose of the reheat cycle:** Reduce the moisture content of the steam at the final stages of the expansion process.





Improvements In The Ideal Rankine Cycle

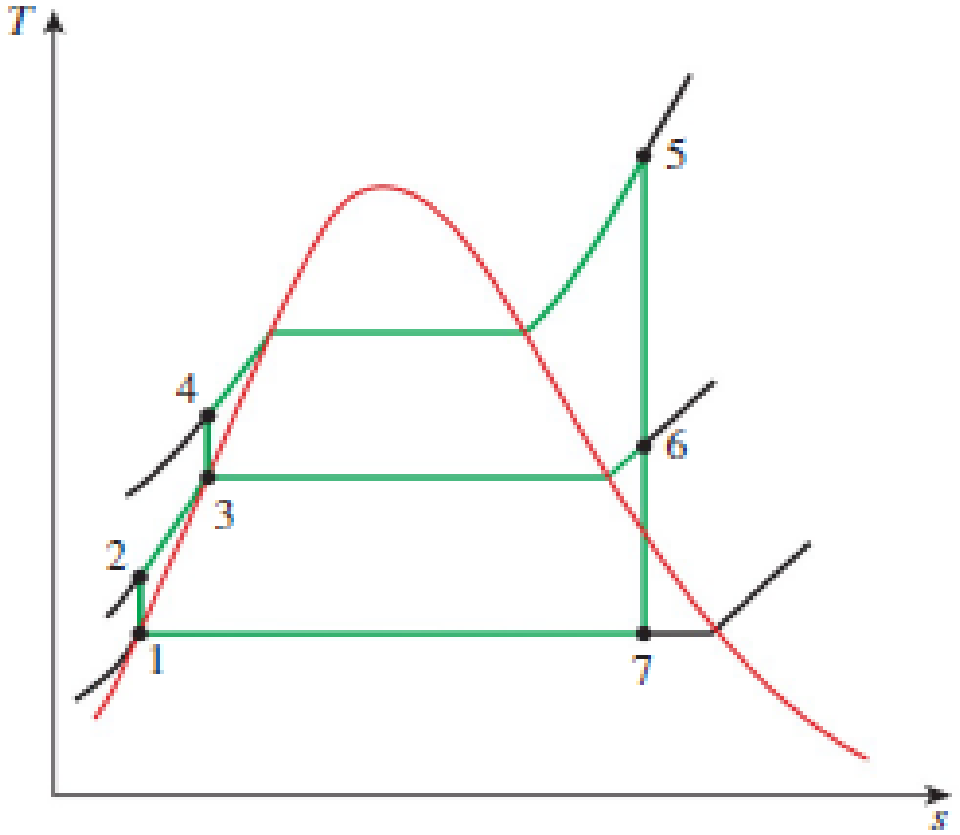
- Modify the simple ideal Rankine cycle with a **regeneration** process.
- **Sole purpose** is to raise the temperature of the liquid leaving the pump (called the **feed water**) before it enters the boiler.
- This reduces the rate at which heat is added to the liquid as it enters the boiler (Thus Q_{in} **reduces**) thereby increasing the cycle efficiency.

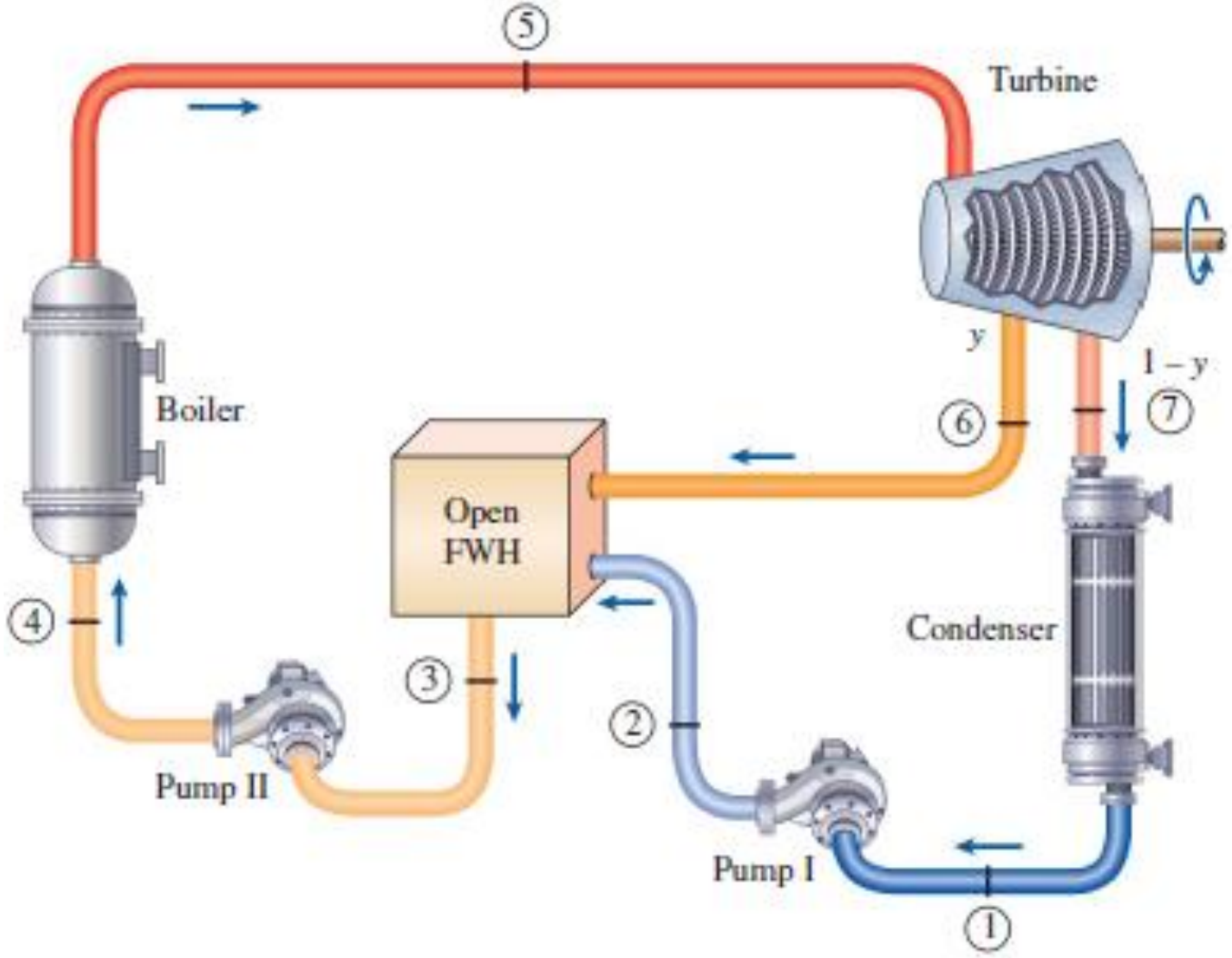
- **Regeneration**
- ✓ Improves cycle efficiency,
- ✓ Provides a convenient means of de-aerating the feed water (removing the air that leaks in at the condenser) to prevent corrosion in the boiler.
- ✓ Helps in controlling the large volume flow rate of the steam at the final stages of the turbine (due to the large specific volumes at low pressures).

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Improvements In The Ideal Rankine Cycle

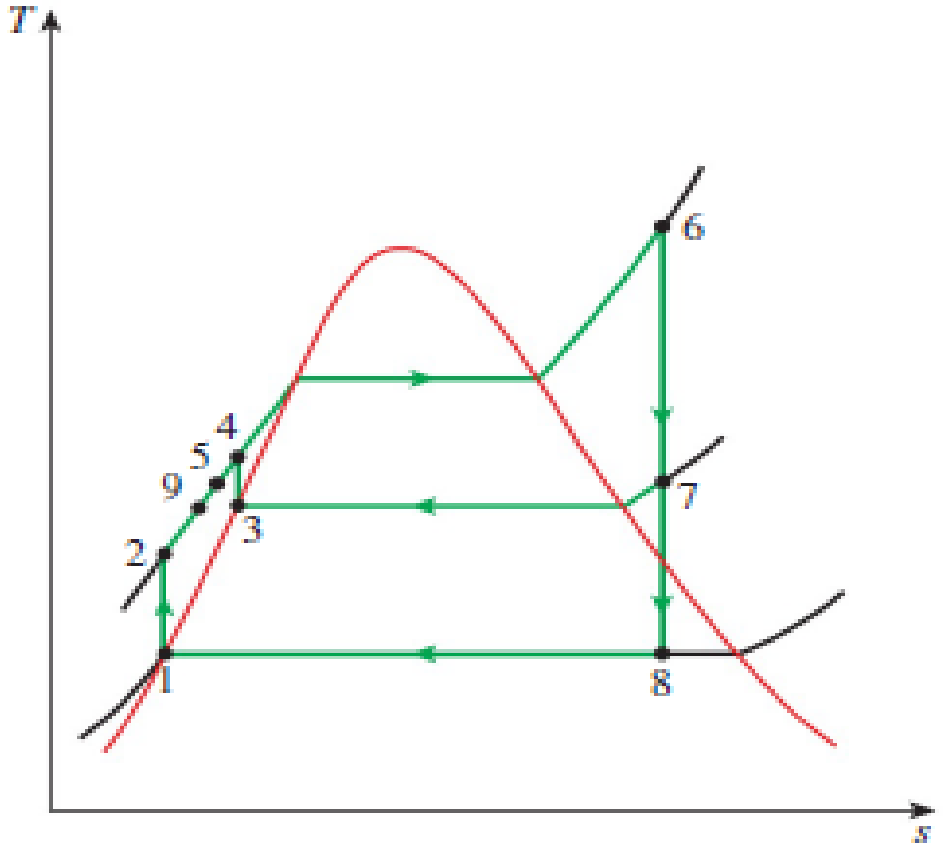
Ideal regenerative Rankine cycle with an open feed water heater

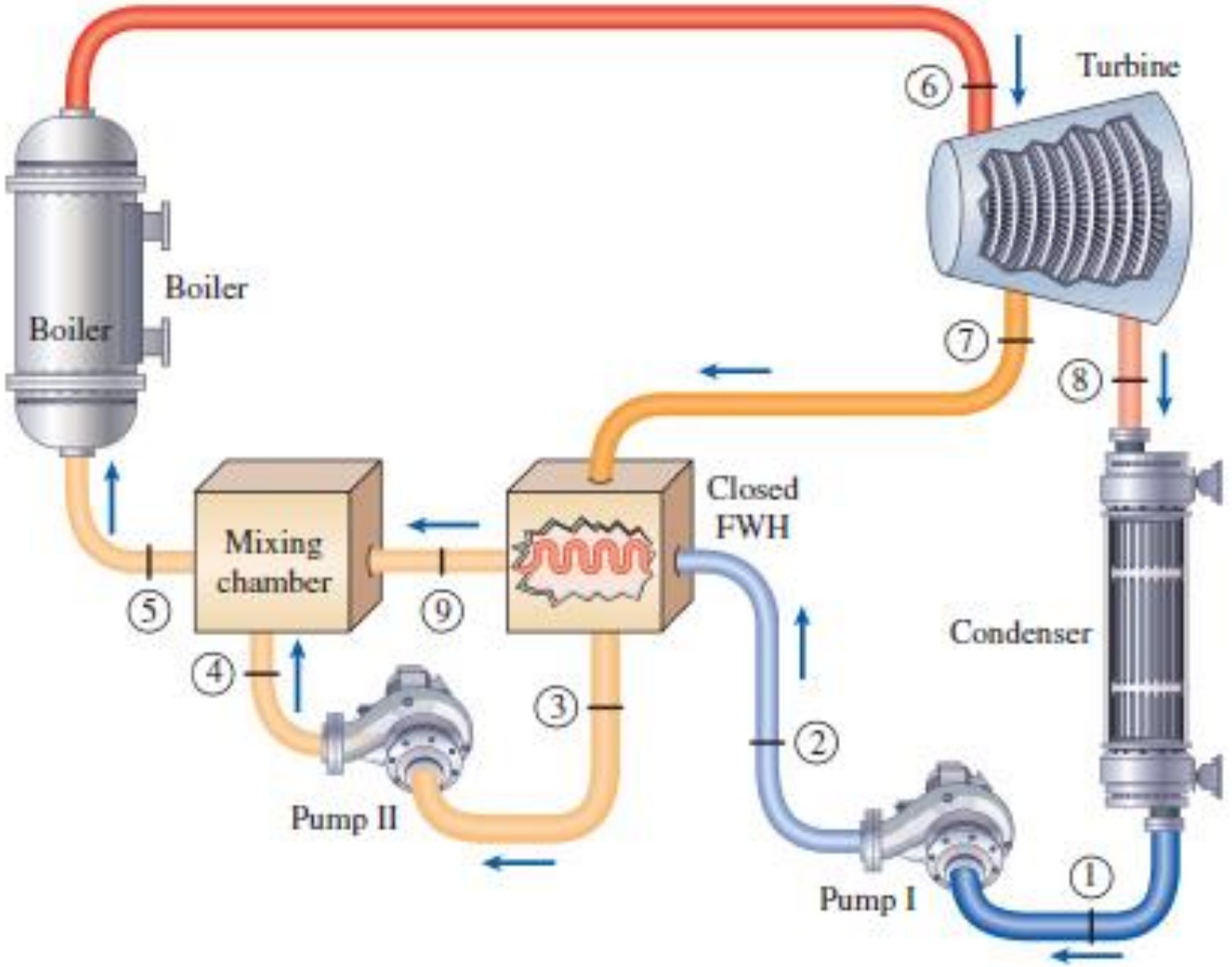




Improvements In The Ideal Rankine Cycle

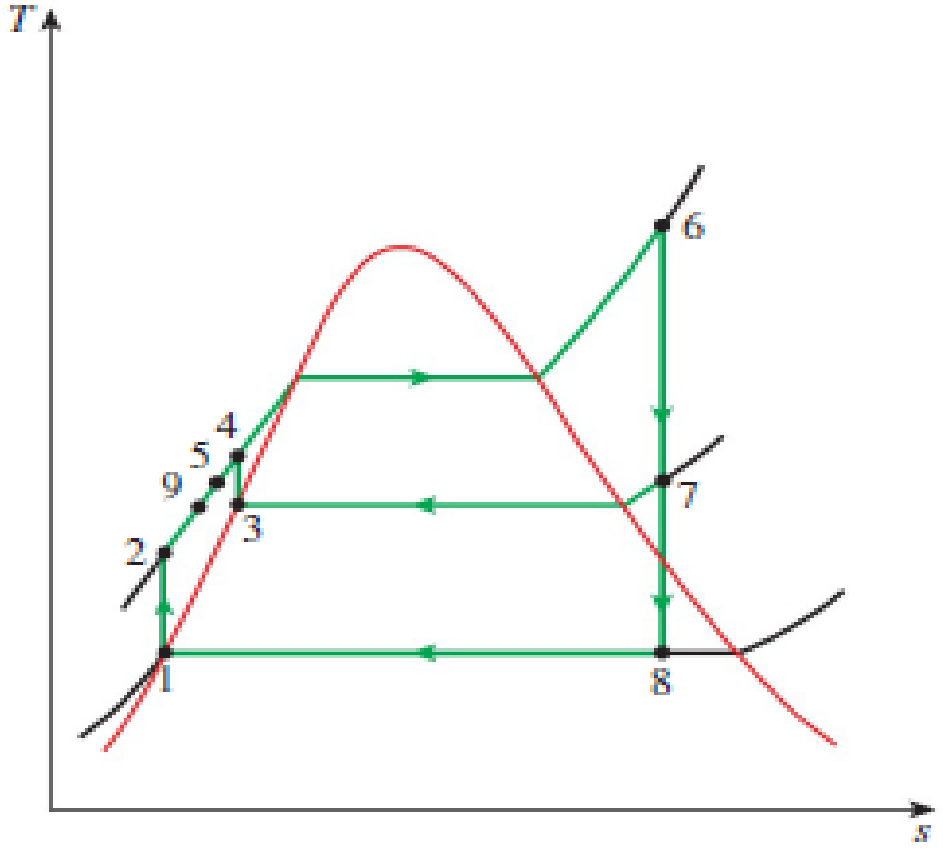
Ideal regenerative Rankine cycle with a closed feed water heater

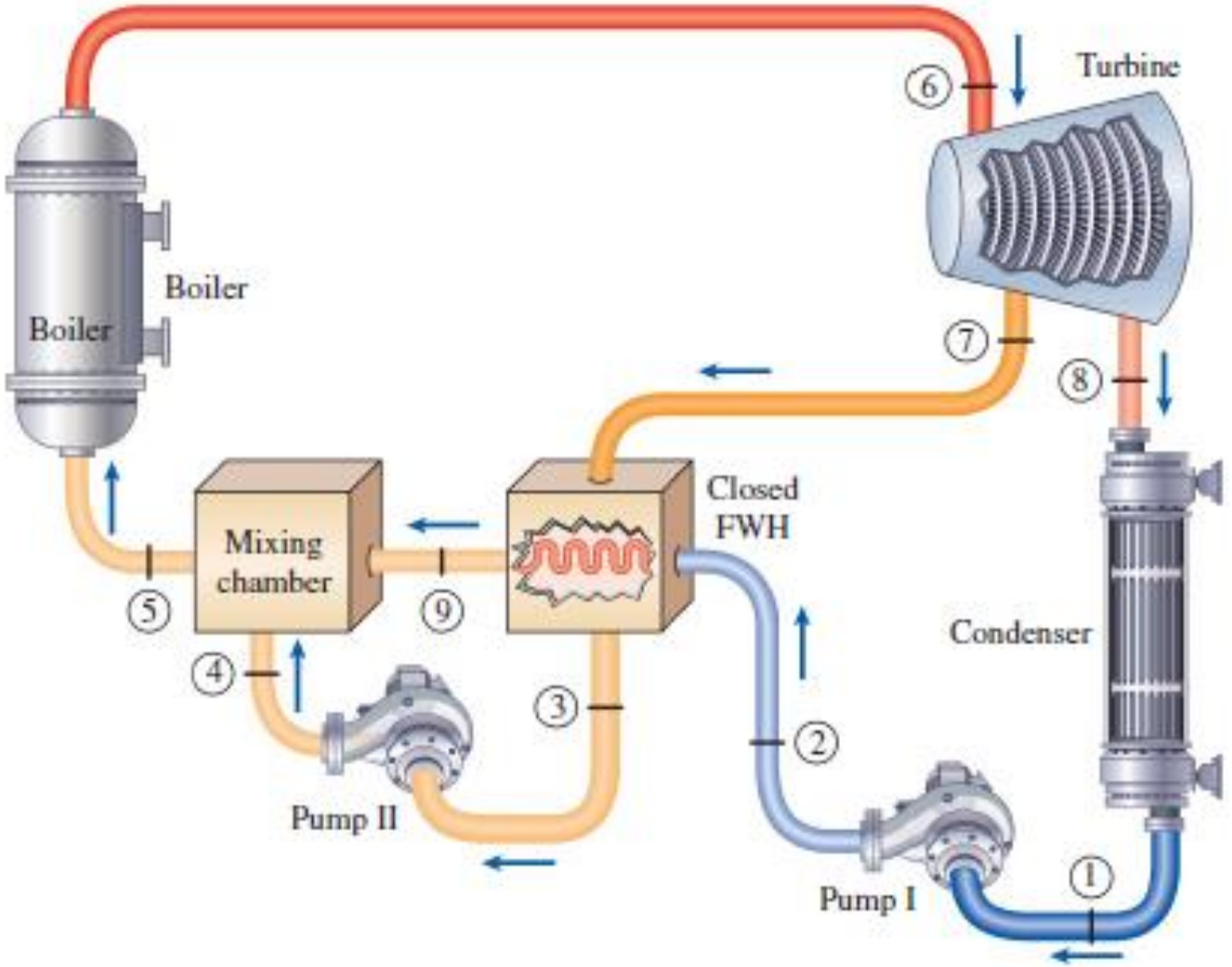




Improvements In The Ideal Rankine Cycle

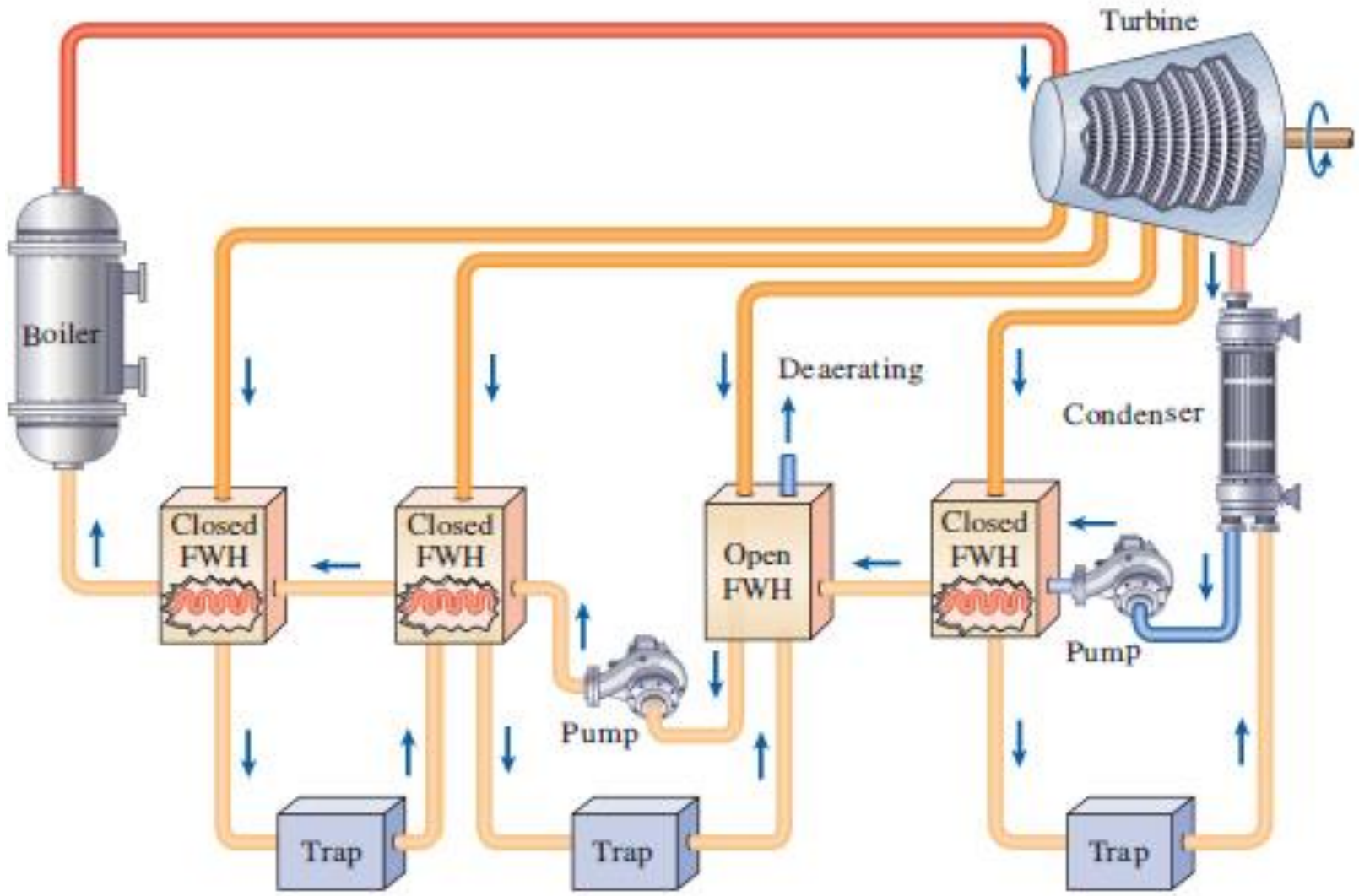
Ideal regenerative Rankine cycle with a closed feed water heater





Improvements In The Ideal Rankine Cycle

A steam power plant with one open and three closed feed water heaters.



Improvements In The Ideal Rankine Cycle

❖ Modify the simple ideal Rankine cycle with a **Cogeneration** process.

- **Cogeneration** is the production of more than one useful form of energy (such as process heat and electric power) from the same energy source.
- **It is appropriate to define a utilization factor ϵ_u for a cogeneration plant as;**

❖ The **utilization factor** of the ideal steam-turbine cogeneration plant is obviously 100 percent.

✓ Actual cogeneration plants have utilization factors as high as 80 percent.

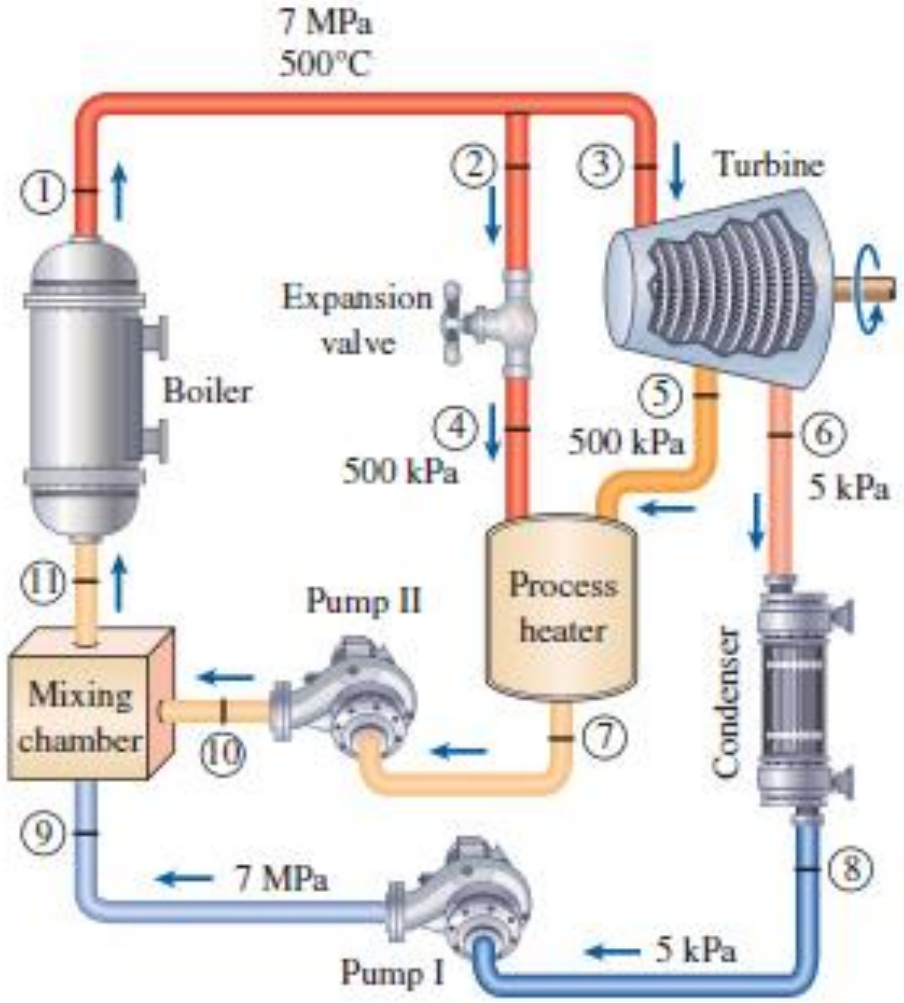
✓ Some recent cogeneration plants have even higher utilization factors.

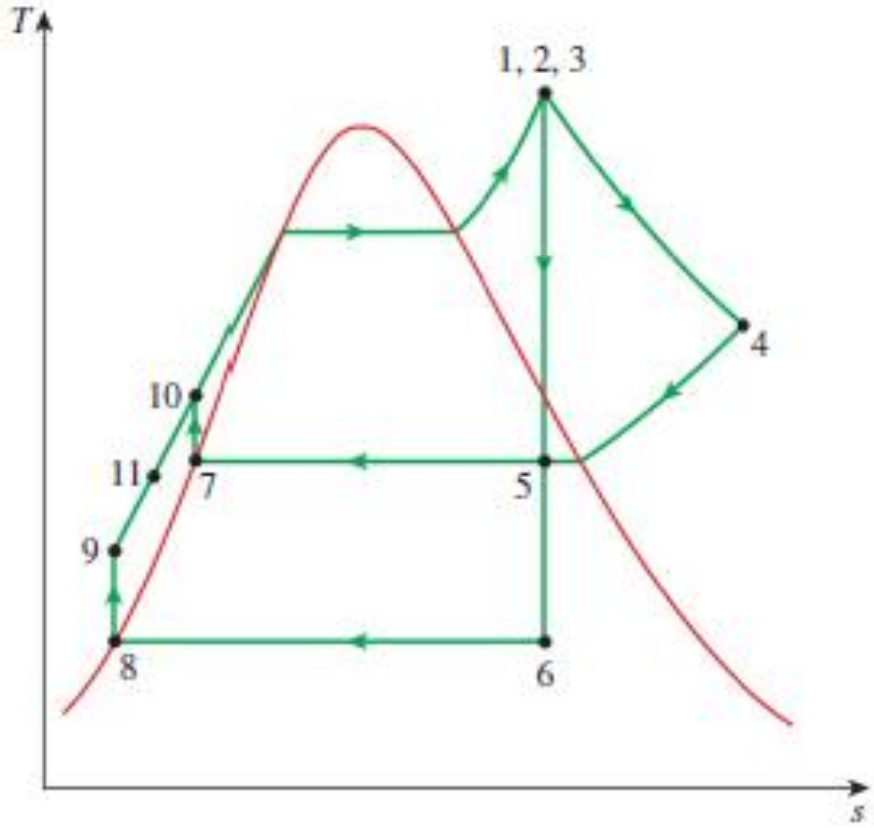
$$\epsilon_u = \frac{\text{Net work output + Process heat delivered}}{\text{Total heat input}} = \frac{\dot{W}_{net} + \dot{Q}_p}{\dot{Q}_{in}} = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}}$$

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Improvements In The Ideal Rankine Cycle

Cogeneration Plant






THANK
YOU

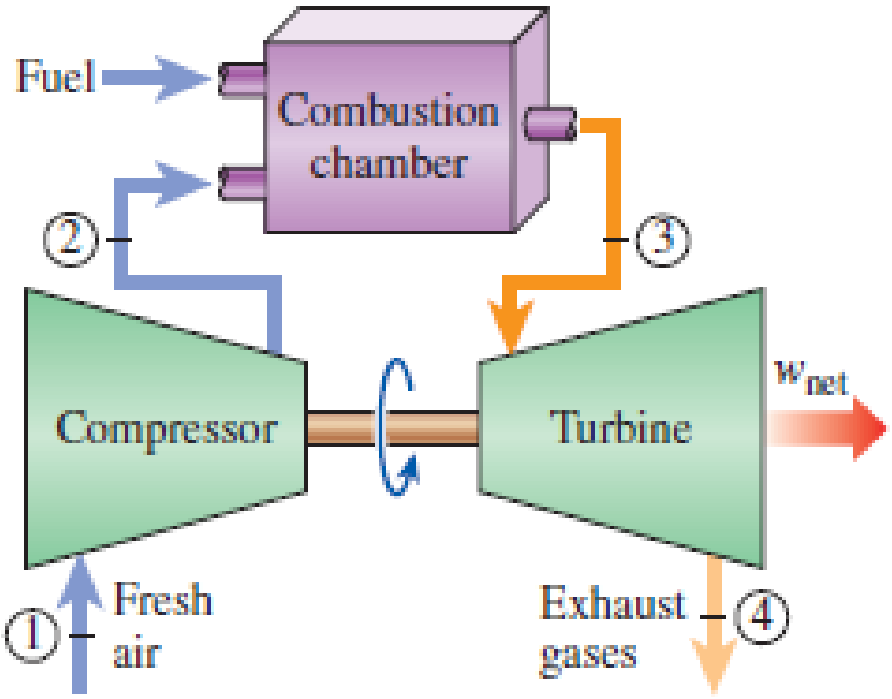
ENGINEERING THERMODYNAMICS (MME 365)

GAS TURBINE POWER CYCLE

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Kwame Nkrumah University of Science and Technology, Kumasi, Ghana

Gas Turbine Power Cycle

- Gas turbines usually operate on an open cycle. Fresh air at ambient conditions is drawn into the compressor, where its temperature and pressure are raised.
- The high pressure air proceeds into the combustion chamber, where the fuel is burned at constant pressure.
- The resulting high-temperature gases then enter the turbine, where they expand to the atmospheric pressure while producing power.
- The exhaust gases leaving the turbine are thrown out (not recirculated), causing the cycle to be classified as an open cycle.

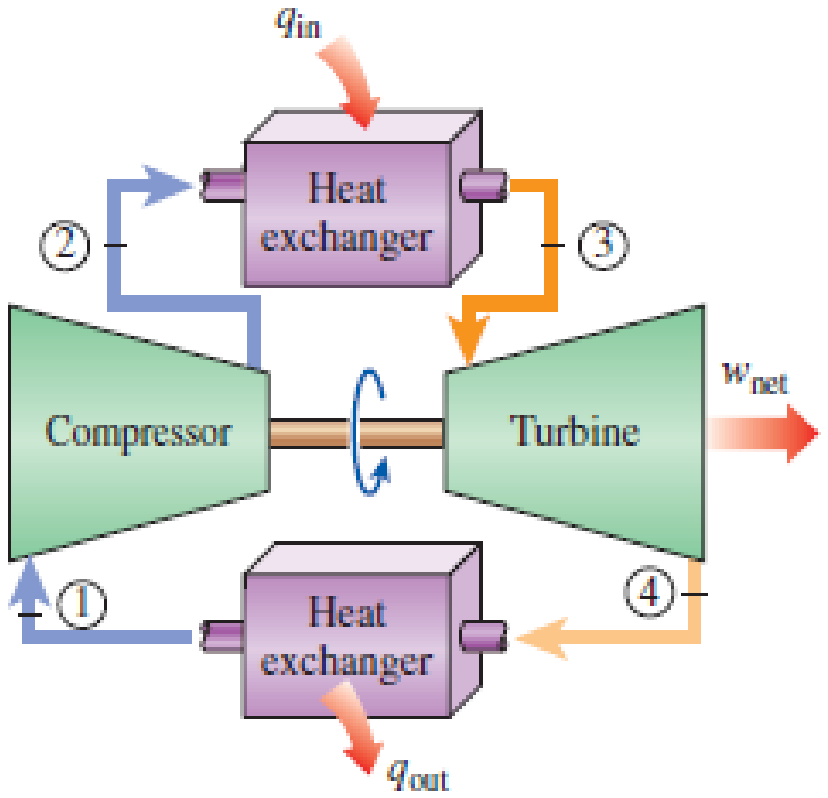


Open cycle Gas power plant

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Gas Power Cycle: Ideal Brayton Cycle

- The open gas-turbine cycle can be modeled as a closed cycle by utilising the **air-standard assumptions.**
- The combustion process is replaced by a constant-pressure heat-addition process from an external source, and the exhaust process is replaced by a constant-pressure heat-rejection process to the ambient air.
- The ideal cycle that the working fluid undergoes in this closed loop is the **Brayton cycle.**



Closed cycle Gas power plant

Gas Power Cycle: Ideal Brayton Cycle

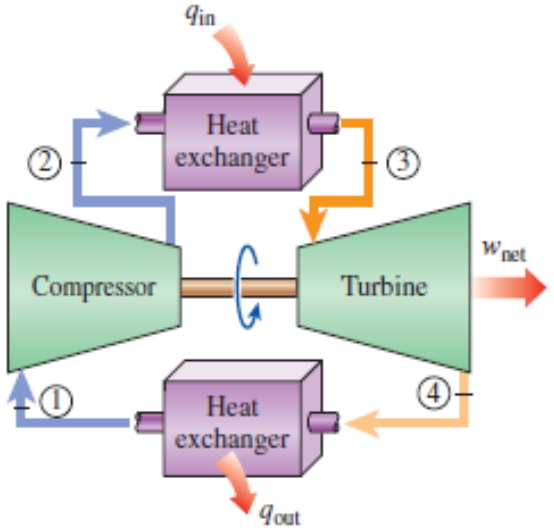
❖ The **ideal Brayton cycle**, is made up of four internally reversible processes:

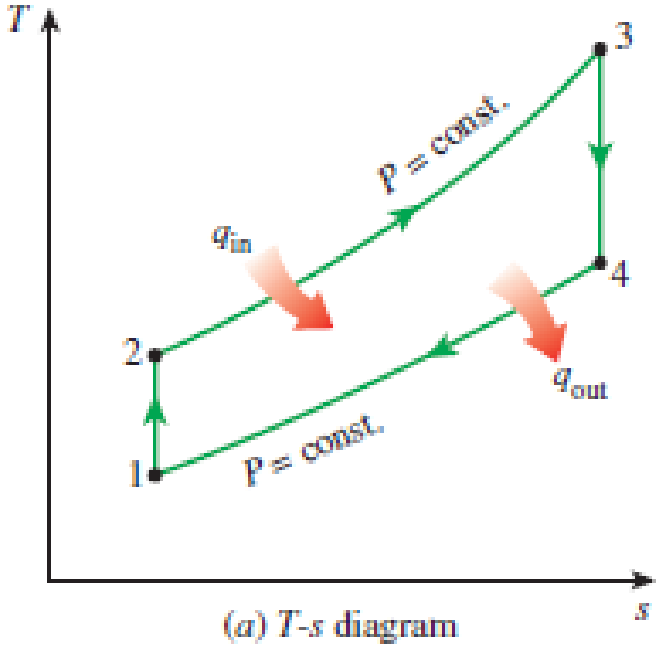
✓ 1-2 Isentropic compression (in a compressor)

✓ 2-3 Constant-pressure heat addition

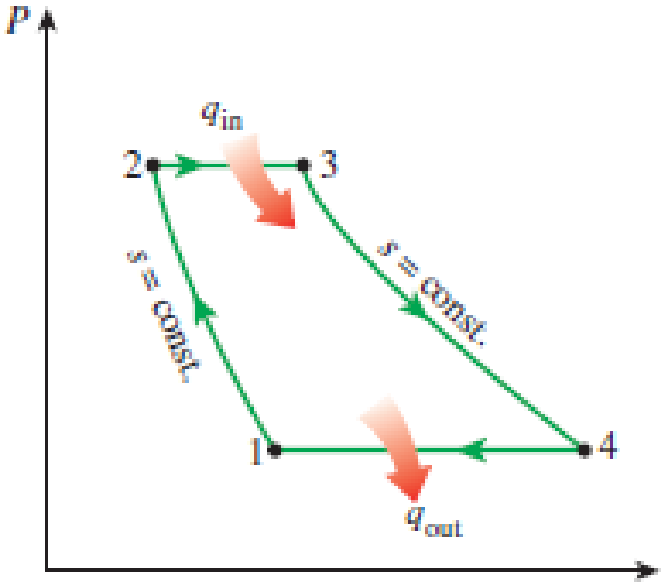
✓ 3-4 Isentropic expansion (in a turbine)

✓ 4-1 Constant-pressure heat rejection





(a) T-s diagram



(b) P-v diagram

Closed cycle Gas power plant

Gas Power Cycle: Ideal Brayton Cycle

$$\dot{Q}_{23} = \dot{m}\,(h_3 - h_2) = \dot{m}\,c_p(T_3 - T_2)$$

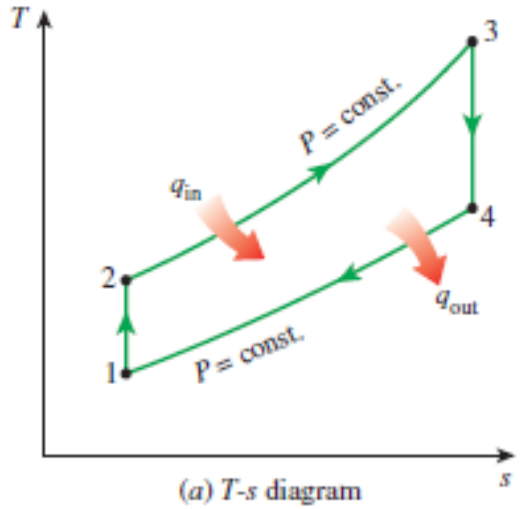
$$\dot{Q}_{41} = \dot{m}\,(h_1 - h_4) = \dot{m}\,c_p(T_1 - T_4)$$

$$\eta_{th} = \frac{|W|}{Q_{in}} = \frac{Q_{net}}{Q_{in}} = \frac{Q_{23} + Q_{41}}{Q_{23}} = 1 + \frac{Q_{41}}{Q_{23}} \qquad \eta_{th} = 1 - \frac{(T_4 - T_1)}{(T_3 - T_2)} = 1 - \frac{T_1}{T_2}\left(\frac{\frac{T_4}{T_1} - 1}{\frac{T_3}{T_2} - 1}\right)$$

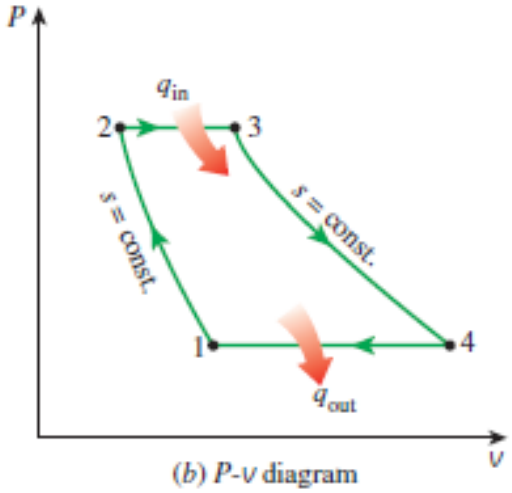
Define $p_2/p_1 = r_p = p_3/p_4$ Also $T_2 = T_1\left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}$ and $T_4 = T_3\left(\frac{p_4}{p_3}\right)^{\frac{\gamma-1}{\gamma}}$

$$\Rightarrow \quad \frac{T_2}{T_1} = \frac{T_3}{T_4} \quad \text{i.e.} \quad \frac{T_4}{T_1} = \frac{T_3}{T_2} \quad \therefore \quad \frac{T_4}{T_1} - 1 = \frac{T_3}{T_2} - 1$$

$$\therefore \quad \eta_{th} = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{r_p^{\frac{\gamma-1}{\gamma}}} = 1 - r_p^{-\frac{\gamma-1}{\gamma}}$$



(a) T-s diagram

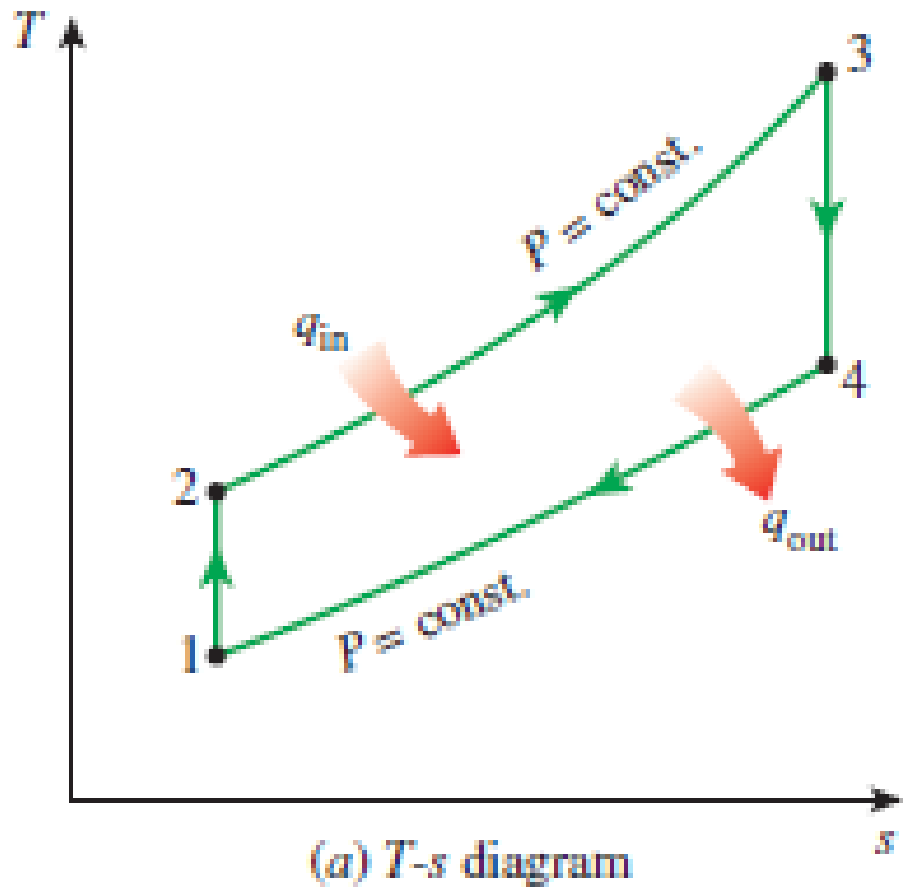


(b) P-v diagram

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Ideal Brayton Cycle Problem

- Consider a simple ideal Brayton cycle with air as the working fluid. The pressure ratio of the cycle is 6, and the minimum and maximum temperatures in the cycle are 300 K and 1300 K, respectively. Now the pressure ratio is doubled without changing the minimum and maximum temperatures in the cycle. Determine the change in
  - ✓ The net work output per unit mass.
  - ✓ The thermal efficiency of the cycle as a result of this modification.
- Take cp for air to be 1.005 kJ/kg.K and γ to be 1.4 kJ/kg. K



(a) T-s diagram

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Ideal Brayton Cycle Problem

Solution: Processes 1-2 and 3-4 are isentropic processes.

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} \Rightarrow T_2 = T_1\left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = (300\,\text{K})(6)^{0.4/1.4} = 500.6\,\text{K}$$

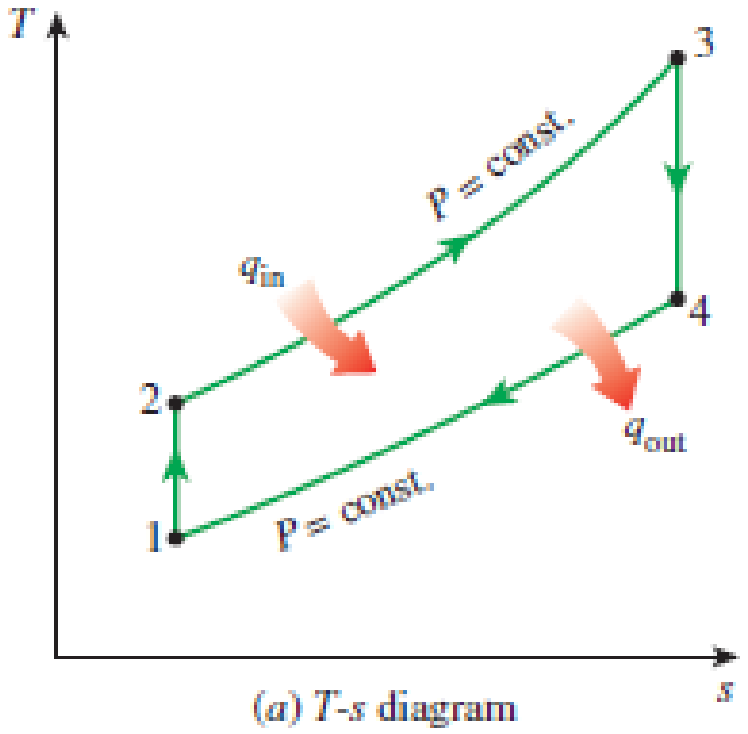
also,
$$T_4 = T_3\left(\frac{P_3}{P_4}\right)^{\frac{\gamma-1}{\gamma}} = (1300\,\text{K})\left(\frac{1}{6}\right)^{0.4/1.4} = 779.1\,\text{K}$$

$$q_{in} = h_3 - h_2 = c_p(T_3 - T_2) = 1.005 \times (1300 - 500.6) = 803.4\,\text{kJ/kg}$$

$$|q_{out}| = h_4 - h_1 = c_p(T_4 - T_1) = 1.005 \times (779.1 - 300) = 481.5\,\text{kJ/kg}$$

$$|w_{net}| = |q_{net}| = q_{in} - |q_{out}| = (803.4 - 481.5)\,\text{kJ/kg} = 321.9\,\text{kJ/kg}$$

$$\text{Thermal efficiency, } \eta_{th} = \frac{|w_{net}|}{q_{in}} \times 100\,\% = \frac{321.9}{803.4} \times 100\,\% = 40.1\%$$



(a) T-s diagram

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Ideal Brayton Cycle Problem

Solution: Considering the effect of doubling the pressure ratio for the cycle i.e. $r_p = 12$

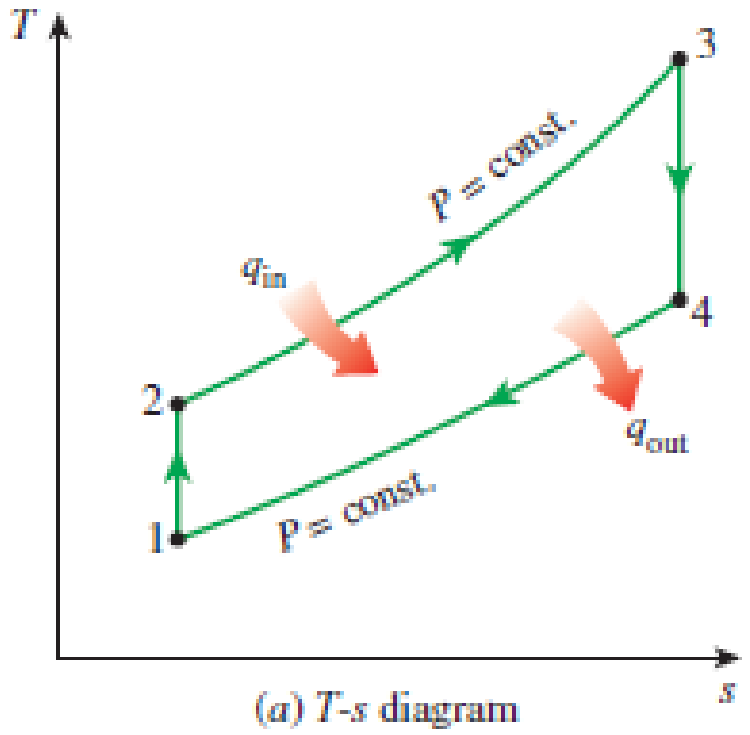
By similar analysis, T_2 and T_4 can be determined as 610.2 K and 639.2 K respectively.

$$\text{and} \quad q_{in} = 1.005(1300 - 610.2) = 693.2\ \text{kJ/kg}$$

$$|q_{out}| = 1.005(639.2 - 300) = 340.9\ \text{kJ/kg}$$

$$|w_{net}| = (693.2 - 340.9) = 352.3\ \text{kJ/kg} \ \text{(increase)}$$

$$\eta_{th} = \frac{352.3}{693.2} \times 100\,\% = 50.8\,\% \ \text{(increase)}$$

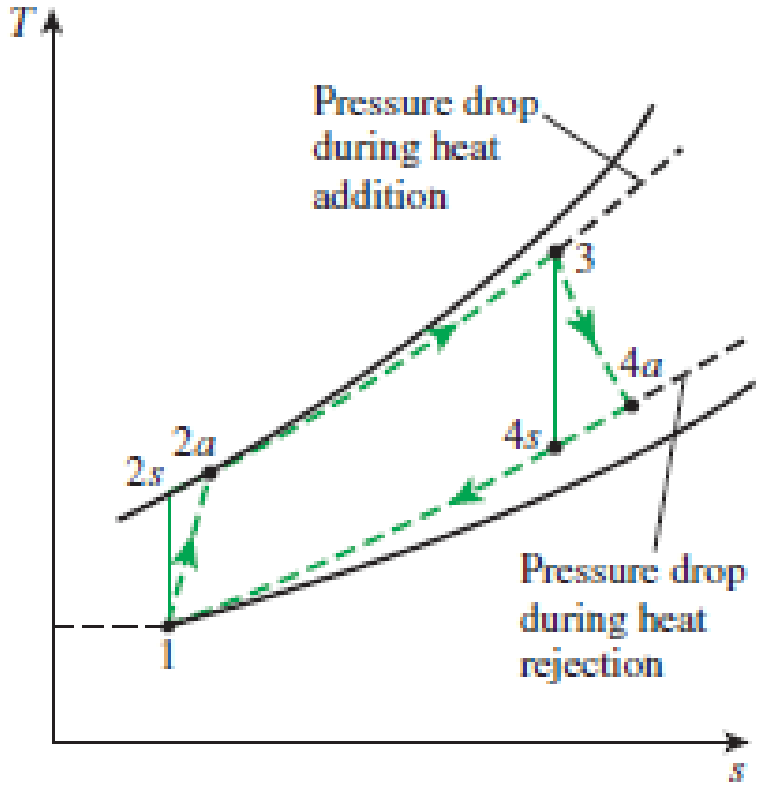


(a) T-s diagram

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Deviation of Actual Gas-Turbine Cycles from Ideal Cycle

- The actual gas-turbine cycle differs from the ideal Brayton cycle on several accounts.
- Pressure drop during the heat-addition and heat-rejection processes is inevitable.
- The actual work input to the compressor is more, and the actual work output from the turbine is less because of irreversibilities.
- The deviation of actual compressor and turbine behavior from the idealized isentropic behavior can be accurately accounted for by utilizing the isentropic efficiencies of the turbine and compressor.



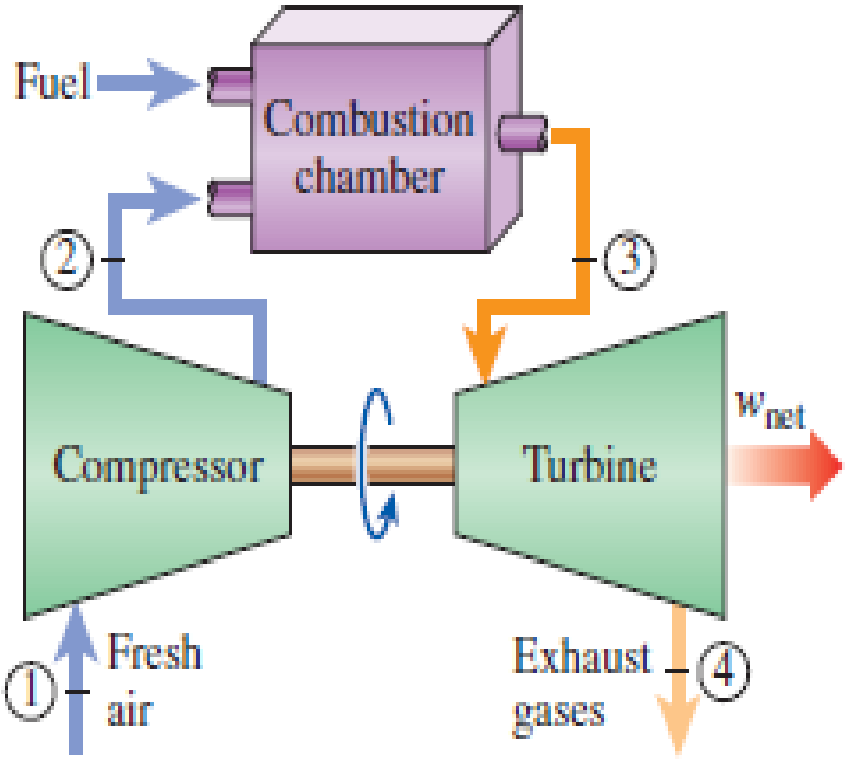
$$\eta_C = \frac{w_s}{w_a} \cong \frac{h_{2s} - h_1}{h_{2a} - h_1}$$

$$\eta_T = \frac{w_a}{w_s} \cong \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$$

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Actual Gas Turbine Cycle Problem

- A gas-turbine power has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Assuming a compressor efficiency of 80% and a turbine efficiency of 85, determine
- ✓ (a) the turbine exit temperature of the gas-turbine cycle.
- ✓ (b) the back work ratio,
- ✓ (c) the thermal efficiency,
- Take cp for air to be 1.005 kJ/kg.K and γ to be 1.4 kJ/kg. K
- ✓ **Suggested Answers: 532 °C; 0.615; 0.274**



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Actual Gas Turbine Cycle Problem

Solution: Processes 1-2s and 3-4s are isentropic

$$T_{2_s} = T_1\left(\frac{P_2}{P_1}\right)^{\gamma-1/\gamma} = 300\ K(8)^{0.4/1.4} = 543.4\ K$$

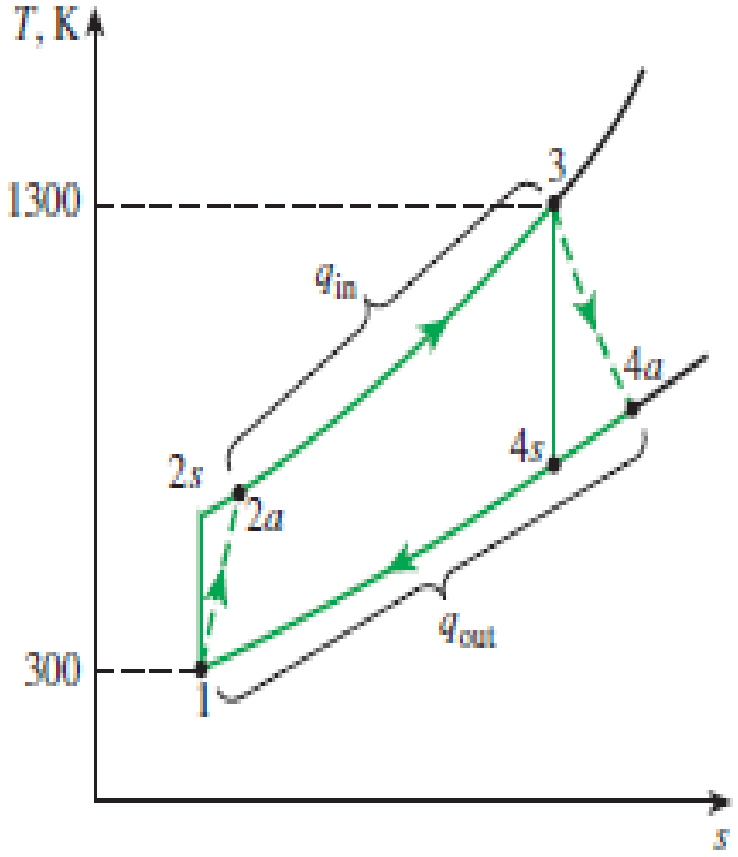
$$\eta_c = \frac{T_{2_s} - T_1}{T_{2_a} - T_1}$$

$$T_{2_a} = \frac{T_{2_s} - T_1}{\eta_c} + T_1 = \frac{543.4 - 300}{0.8} + 300\ K = 604.25\ K$$

$$T_{4_s} = T_3\left(\frac{P_4}{P_3}\right)^{\gamma-1/\gamma} = 1300\ K\left(\frac{1}{8}\right)^{0.4/1.4} = 717.7\ K$$

$$\eta_t = \frac{T_{4_a} - T_3}{T_{4_s} - T_3};$$

$$T_{4_a} = \eta_t\left(T_{4_s} - T_3\right) + T_3 = 0.85(717.7 - 1300) + 1300 = 805.05\ K$$



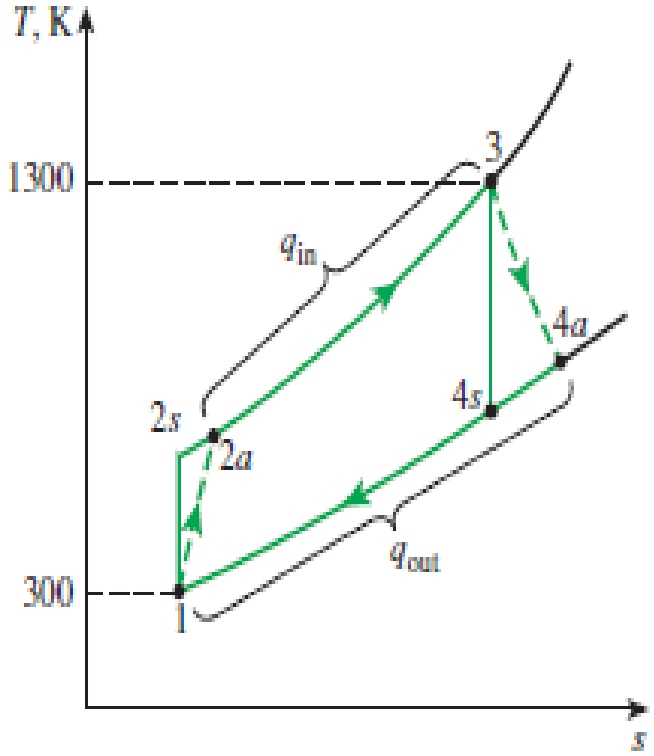
Actual Gas Turbine Cycle Problem

Solution:

$$BWR = \frac{|W_{Compressor}|}{|W_{turbine}|} = \frac{T_{2_a} - T_1}{T_3 - T_{4_a}} = \frac{604.25 - 300}{1300 - 805.05} = 0.615$$

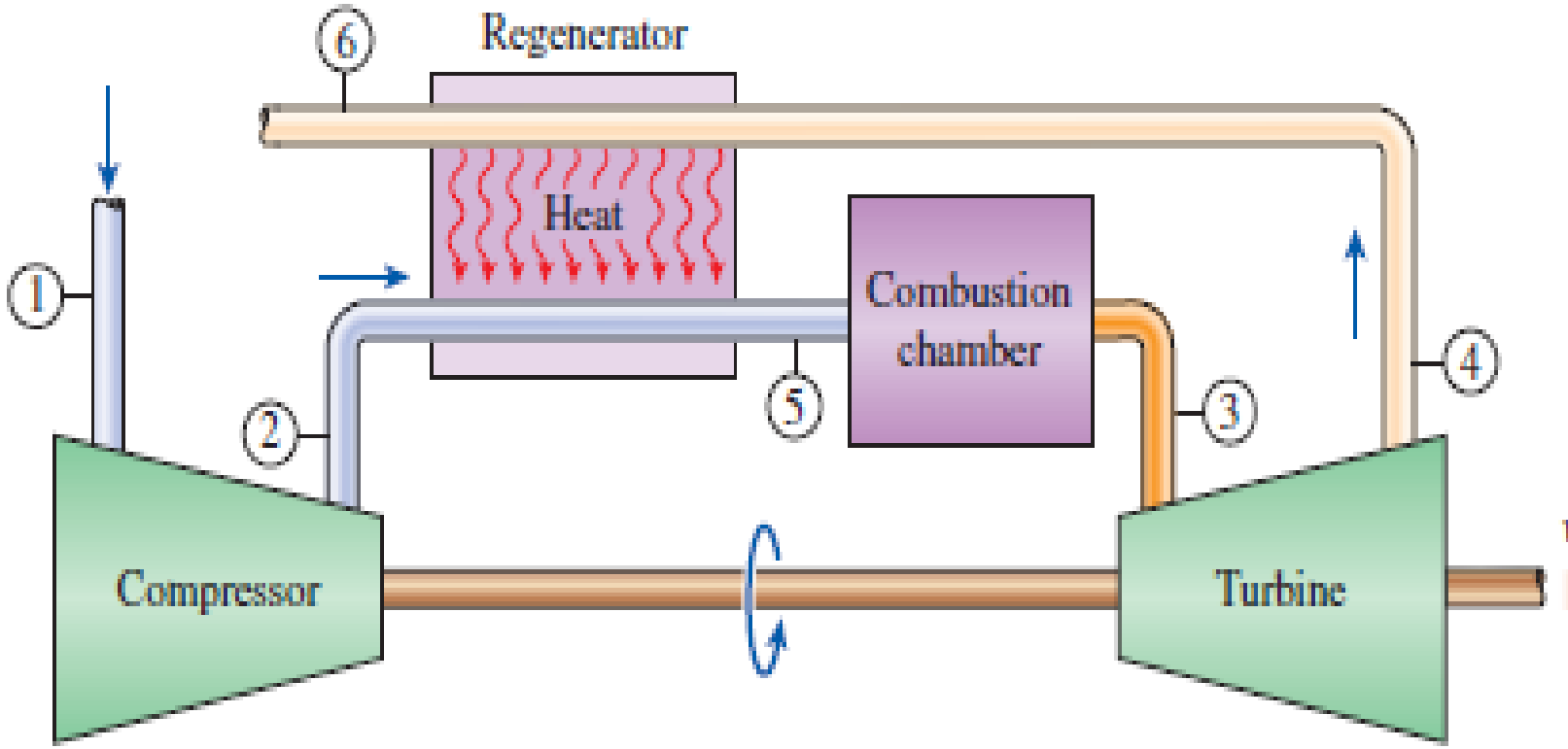
$$\eta_{cycle} = \frac{W_{net}}{Q_{in}} = \frac{Q_{in} + Q_{out}}{Q_{in}} = 1 + \frac{c_p(T_1 - T_{4_a})}{c_p(T_3 - T_{2_a})}$$

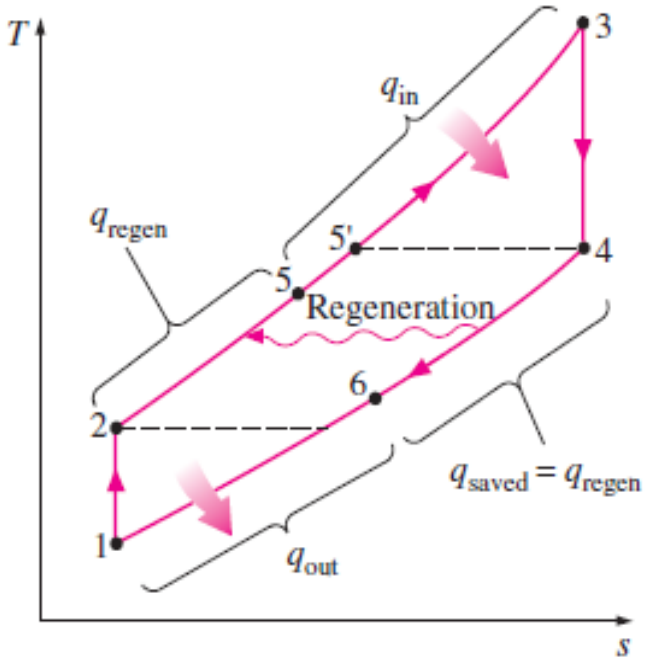
$$\eta_{cycle} = 1 + \frac{(300 - 805.05)}{(1300 - 604.25)} = 0.274$$



Improvements In The Gas Turbine Cycle

- Using a **Regenerator** to increase the temperature of the air entering the combustion chamber.



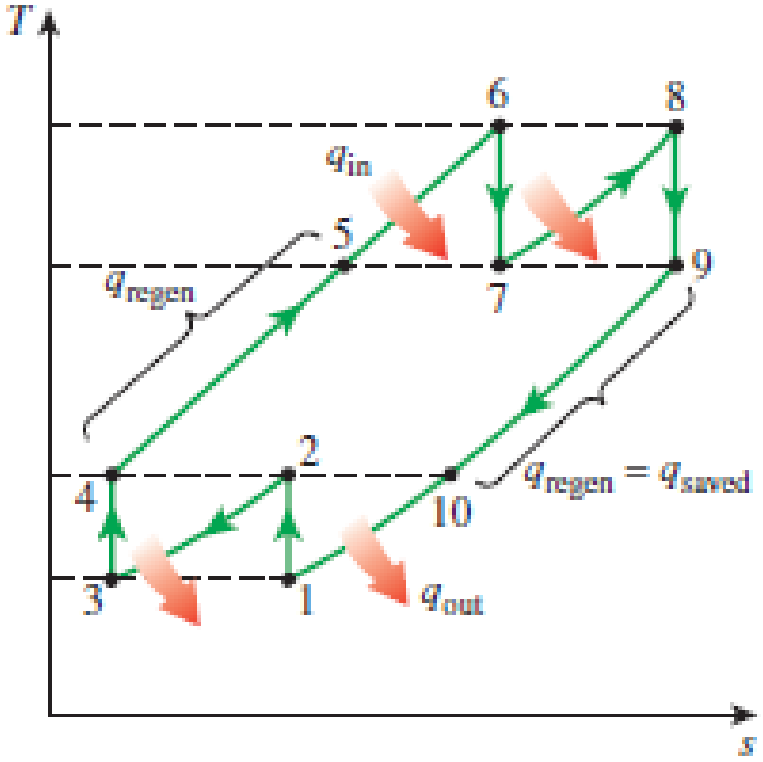


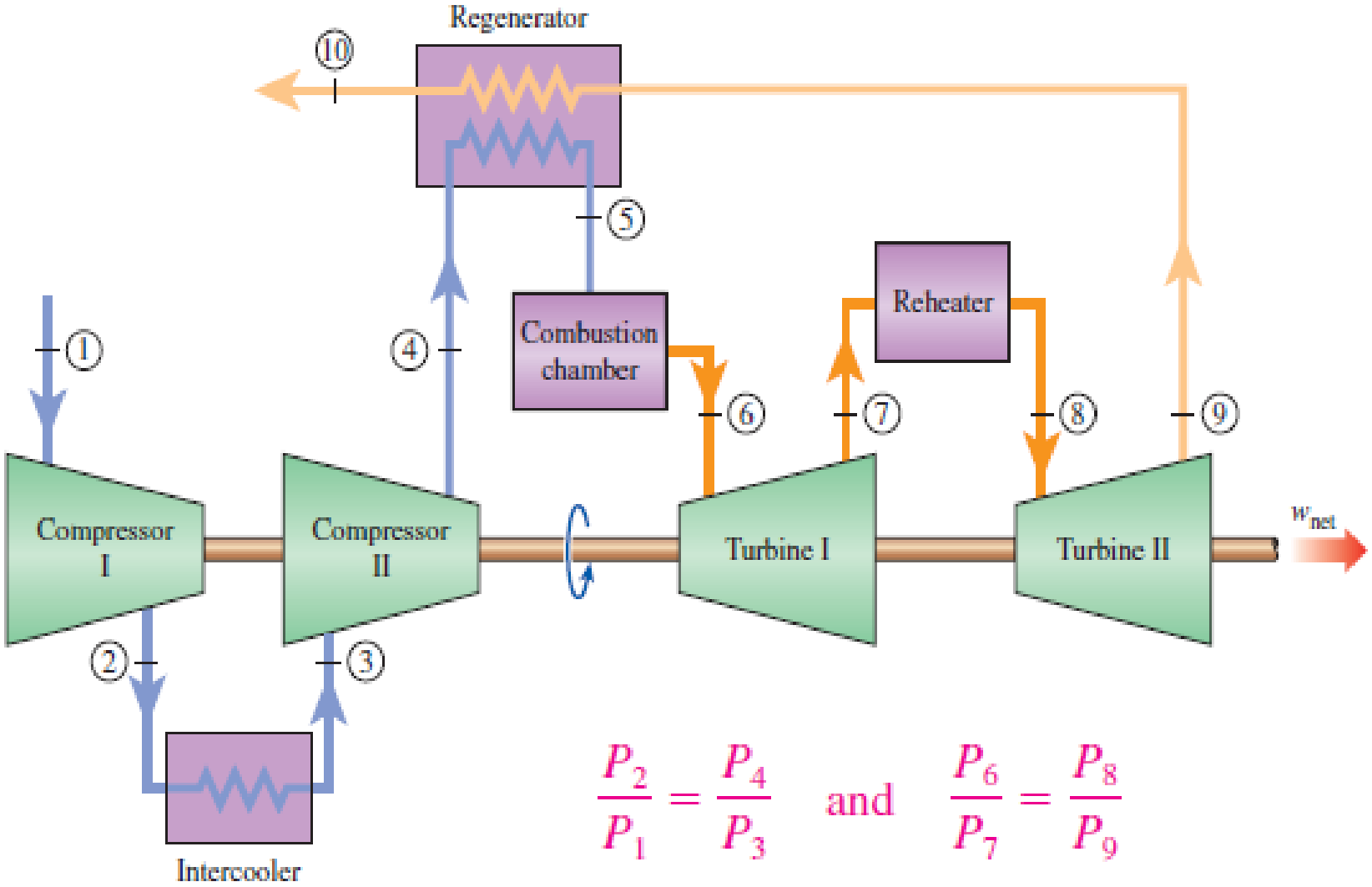
$$\epsilon = \frac{q_{regen,act}}{q_{regen,max}} = \frac{h_5 - h_2}{h_4 - h_2}$$

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Improvements In The Gas Turbine Cycle

- Using
  - ✓ Regenerator
  - ✓ Intercooler
  - ✓ Reheater





$$\frac{P_2}{P_1} = \frac{P_4}{P_3} \quad \text{and} \quad \frac{P_6}{P_7} = \frac{P_8}{P_9}$$


THANK
YOU